REVIEW

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

Focusing on individualized nutrition within the algorithmic diet: an in-depth look at recent advances in nutritional science, microbial diversity studies, and human health

Reyed M Reyed[1*]

[1]Bioprocess Development Department, Genetic Engineering and Biotechnology Research Institute (GEBRI), City for Scientific Research and Technological Applications ''SRTA-CITY'', Alexandria 21934, Egypt.

[*]**Corresponding to:** Reyed M Reyed, Bioprocess Development Department, Genetic Engineering and Biotechnology Research Institute (GEBRI), City for Scientific Research and Technological Applications ''SRTA-CITY'', North Coast Road at the 50th kilometer, New Borg El-Arab City, Alexandria 21934, Egypt. E-mail: nutritionalgutmicrobiome@internet.ru.

Competing interests

The author declares no conflicts of interest.

Acknowledgments

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Abbreviations

MS, mass spectroscopy; FDBs, food distribution databases; FCD, food composition data; FCDB, food composition database; FPIP, food platform regarding industrial processing; FAO, food and agriculture organization; LCI, life cycle inventories; LCA, life cycle analysis; WES, whole-exome sequencing; GWAS, genome-wide association studies; GOC, gene ontology consortium; CBA, computer-based analysis; JGI, joint genome institute; LBNL, lawrence berkeley national laboratory; EMSL, environmental molecular systems laboratory; PNNL, pacific northwest national laboratory.

Citation

Reyed RM. Focusing on individualized nutrition within the algorithmic diet: an in-depth look at recent advances in nutritional science, microbial diversity studies, and human health. *Food Health*. 2023;5(1):5. doi: 10.53388/FH2023005.

Executive editor: Na Liu.

Received: 03 November 2022; **Accepted:** 27 January 2023; **Available online:** 01 February 2023.

Abstract

A healthy balanced diet and a healthy lifestyle are very closely linked. Whichever the biological link is, it is overwhelming to understand. Modifications in how food is served, divided up, and supervised, such as the introduction of nutritional hygiene standards, food handling practices, and the entry of macro and micronutrients, have had a big impact on human health in the last few decades. Growing evidence indicates that our gut microbiota may affect our health in ways that are at least in part influenced by our diet and the ingredients used in the preparation of our food and drinks, as well as other factors. As a new problem, this one is getting a lot of attention, but it would be hard to figure out how the gut microbiota and nutrition molecules work together and how they work in certain situations. Genetic analysis, metagenomic characterization, configuration analysis of foodstuffs, and the shift to digital health information have provided massive amounts of data that might be useful in tackling this problem. Machine learning and deep learning methods will be employed extensively as part of this research in order to blend complicated data frames and extract crucial information that will be capable of exposing and grasping the incredibly delicate links that prevail between diet, gut microbiome, and overall wellbeing. Nutrition, well-being, and gut microorganisms are a few subjects covered in this field. It takes into account not only databases and high-speed technology, but also virtual machine problem-solving skills, intangible assets, and laws. This is how it works: Computer vision, data mining, and analytics are all discussed extensively in this study piece. We also point out limitations in existing methodologies and new situations that discovered in the context of current scientific knowledge in the decades to come. We also provide background on "bioinformatics" algorithms; recent developments may seem to herald a revolution in clinical research, pushing traditional techniques to the sidelines. Furthermore, their true potential rests in their ability to work in conjunction with, rather than as a substitute for, traditional research hypotheses and procedures. When new metadata propositions are made by focusing on easily understandable frameworks, they will always need to be rigorously validated and brought into question. Because of the huge datasets available, assumption analysis may be used to complement rather than a substitute for more conventional concept-driven scientific investigation. It is only by employing all of us that we will all increase the quality of evidence-based practice.

Keywords: nutritional databank; nutritional human gut microbiota; microbiome; big data; algorithms; public health; dietary programming application

Background

When artificial intelligence technology came out in the fifth generation, it revealed a lot about these small-sized organisms, which have a big impact on the human race, from when they affect sperm and get into the womb, to when they come out during childbirth and get in contact with vaginal microbes, to when they grow up and become adults, and so on. If you think about it, it's a very complicated case. There are several variables at play here. According to poet Samuel Taylor Coleridge, there are "secret universes inside universes" that may be located in the inner gastrointestinal microbiota. There are upwards of 1,000 microbial communities and millions of genes living among the trillions of microorganisms that make up our internal organs. Each person indeed has a different intestinal microbiota, which makes it even more mysterious how our bodies' tiny ecosystems work. Even though scientific researchers have been studying human gut microbes for centuries, it is only lately that they have begun to analyse the multifaceted link between intestinal microbiota, food, homeostasis, and long-term illness. In the past few generations, microbiologists were confined to examining what could be grown in the laboratory. Scientists may investigate microbes without cultivating them in a flask in the lab because of innovations in computational methods (e.g., 16S rRNA gene work, general progress in DNA-based sequencers, etc.). The Fifth Industrial Revolution of AI has changed a lot over the years in our understanding of what microbes are in the human gut microbiota "microbiome" a complex superorganism that can affect physical wellbeing and autoimmune disease [1, 2]. It's hard for us to understand everything about the human microbiome because we don't know how it works with the host and the environment. But new findings have opened up new possibilities in nutrition and dietetics, bioengineering, precision medicine, and other disciplines [3, 4]. While next-generation sequencing and computational biology have made it easier to collect and analyse biochemical and molecular data, they have also made it easier to convert the data into information [5, 8]. Since there have been a lot of new things that have been done in this area, the integration of diet and health data could lead to a new way of measuring well-being, treating illnesses, and designing products. If we want to use this power, we need to gain more knowledge about how to change the proportion and enzymatic dynamics of microbial communities to achieve desired long-term functional outcomes, from lifestyle changes and genetically enhanced offerings [9], to develop more effective improvements and better life quality with a specific focus on food, gastrointestinal tract microbiota, health data, and computational techniques Figure 1 [10]. This article describes the study on the subject of differential geometry mental health that has been carried out. As we get started, we'll take a short look at what we do know and where we can go from here in terms of future studies on the relationship between foodstuffs, general wellbeing, and intestinal microbiota. After that, we'll look at the current datasets and algorithms that are being used, as well as the potential role of artificial intelligence technology [11]. We'll wrap things up with a look at the current state of intangible asset legislation and regulation. Finally, we make recommendations on how to better integrate research resources and technology, particularly within the context of combinatorial technology and numerical modelling [12], to minimise the time it takes to conduct research and development and thereby save money. When it comes to applying high-performance computers to the study of colonic microbiota, dietary habits, and people's lives, metagenomic virtualization methods and bioinformatics [13], and interpretation give little advice for our work. Consequently, by drawing on existing research, outlining important hurdles and potential solutions, and proposing a framework for empirical investigations, the objective of this review is to assist in bridging unresolved questions in this fascinating and potentially ground-breaking topic area. It will achieve significant results by covering key concepts, outlining major challenges and viable solutions, and proposing a mechanism to increase the likelihood of common tasks to accelerate progress in this exciting and potentially innovative field.

Introduction

Understanding GI tract microbiota role in population health

The microbiome has developed as a natural ecosystem that can affect wellbeing and disorder throughout the last twenty years [14]. Recently discovered aspects of the gut genome, notwithstanding our poor knowledge of how it interacts with both the human body and its ecosystem, have opened up new frontiers in subjects such as nutrition and dietetics, targeted therapies, and bioengineering, among other domains [15, 16]. As a result of recent advancements in genomic information and computational biology, low-cost technologies for collecting biological and clinical data [17], have become available, as have tools for converting the data into valuable information. A combination of data from nutrition, gut microbiota, health, and the environment can trigger a framework change in the way wellness states are defined, illnesses are evaluated, items are manufactured, and health treatments are delivered. Progress in understanding is necessary to boost the composition and metabolic dynamics of microbial communities in connection to targeted long-term functional goals, which might range from dietary interventions and biopharmaceutical ingredients to dietary modification and the home environment, see Figure 2 [18-20]. Throughout this research report, we provide an overview of the data that have been collected in the area of food, gut microbiota, human health data, and multivariate statistics. We begin with a quick assessment of the important areas of existing understanding about the interplay between nutritious food, life quality, and the bowel microbiome are all interconnected, followed by a discussion of future directions for research. After that, we'll go through the data sources that are now accessible and the computational techniques that are currently being employed, as well as the potential role that machine learning and artificial intelligence may play in this field, before summarizing the intellectual property "Product innovation" and legislative and regulatory environment. We conclude with proposals for accelerating research and development efforts via improved integration of research resources and technologies, particularly in the case of supercomputing knowledge and numerical modeling.

Knowledge base pipeline

Metagenomic data processing approaches reviewed in recent studies on computational tools for Microbiome data give minimal advice on applying artificial intelligence and data mining [21], that does not provide constructive comments to lifestyle, intestinal microbiota, and functional human clinical information [22]. This is a major problem. The objective of this publication is to completely bolster this vacuum by analyzing academic articles, highlighting significant difficulties and alternative responses, and offering a framework to increase the possibility for research initiatives to expedite development in this fascinating and possibly transformational topic. A growing body of research demonstrates that the human Microbiome knows how to play an important role in human nutrition, how people eat, how they act, and personality characteristics [23]. Prenatal transmission of intestinal microbiota seeds has been shown in several studies, as have variations in childbirth delivery methods in terms of gut microbiota makeup. Babies born by vaginal delivery have a lot of microbes in their digestive tracts from their mother, but babies born by cesarean are more likely to have microbes from their mother's skin and places where they come into contact with other people [24, 25]. Early childhood is a time of rapid changes in the Microbiome's composition. Early childhood is a time of rapid changes in the Microbiome's composition. Throughout the first three to five years of childhood, the Microbiome develops to closely resemble that of an adult. Newborns have a variety of microbiota in their digestive systems after eating various foods [26]. This is becoming more and more clear probiotics Lactobacillus [27], and Bifidobacterium [28-35], are more common in babies who are breastfed, but Clostridium difficile spores, Enterobacteria, and Granulicatella adiacens are more common in babies who are fed formula [36, 37]. The diversity of bifidobacteria in newborns who are exclusively breastfed is also much higher than in

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

toddlers who are fed formula. Multiple studies have shown that these bacterial communities are linked to good health [38-42]. People who have allergies to commercial products [43], overweight [44], and bronchitis [45], all have microbes in their digestive tracts, which have been linked to several long-term illnesses. There are a variety of factors that may influence the makeup of a person's gut microbiota, including antimicrobial medication, changes in a person's long-term diet, microbial illnesses Figure 2 [46, 47], and the way he or she conducts his or her daily life. Individuals who suffer from these illnesses are more susceptible to having difficulties with their gastrointestinal microbiota. If a person has Crohn's disease and other conditions that make it more likely that they will get pathogenic microbes in their gut, they are also more likely to get harmful microorganisms in the digestive tract, too [48-51].

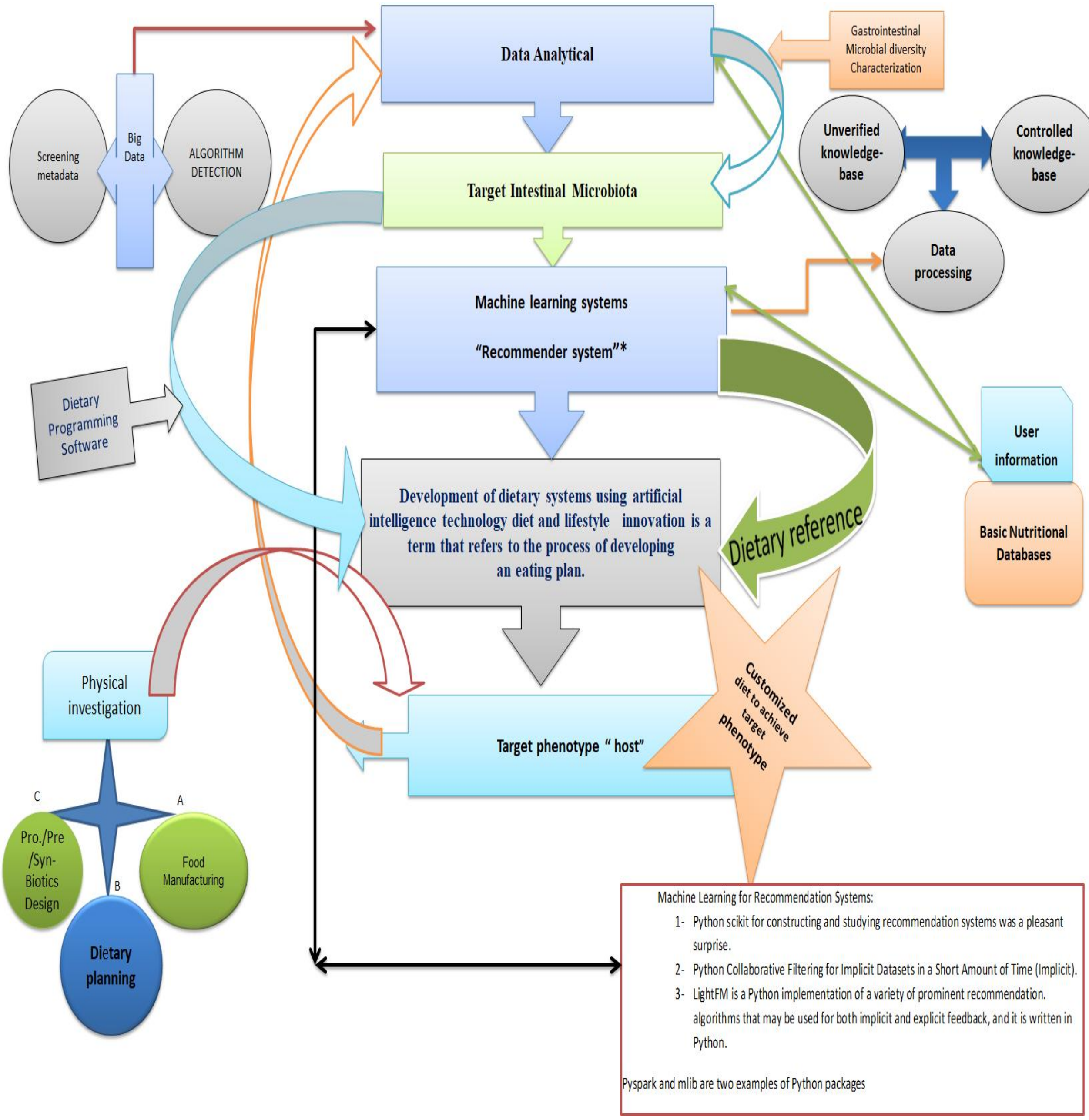

Figure 1 Nutritional management and commercial marketing will strengthen the coming worldwide dietary revolution. In nutrition and dietetics omics techniques, computational modelling is used to extract useful information from raw, high-throughput sequencing data. A dietary suggestion system helps people achieve their ideal microbiome. A nutrition database and people's profiles will be combined to find trends.

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

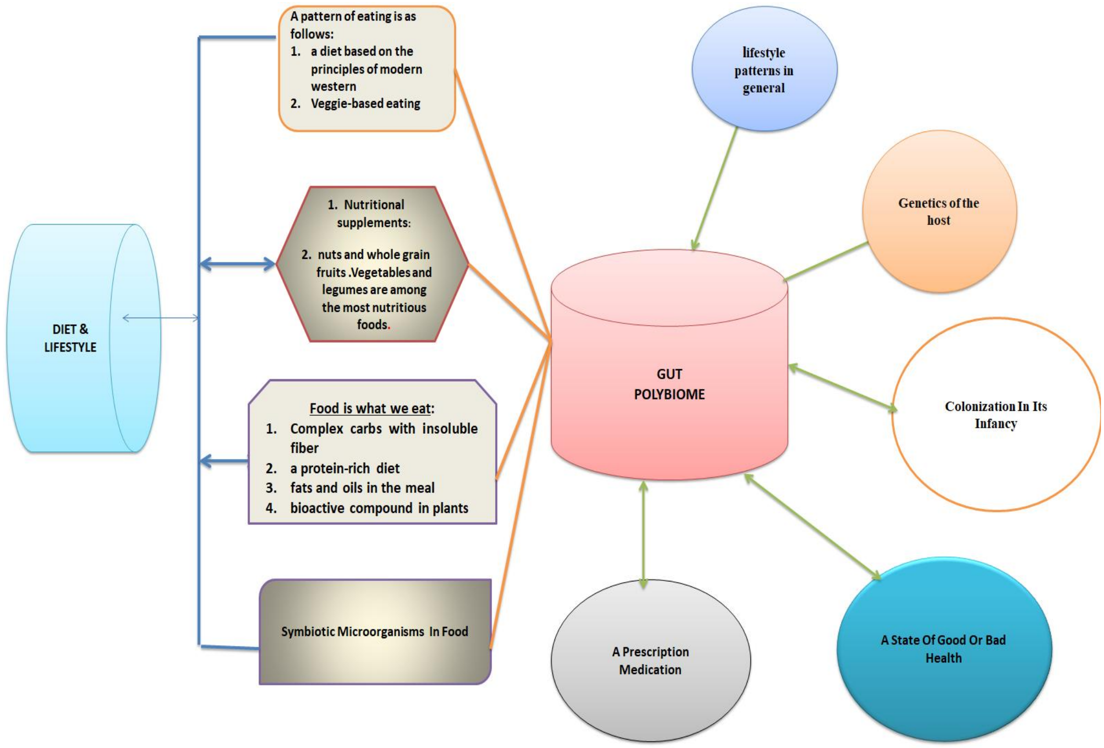

Figure 2 Different environmental factors influence and affect the gut microbiota

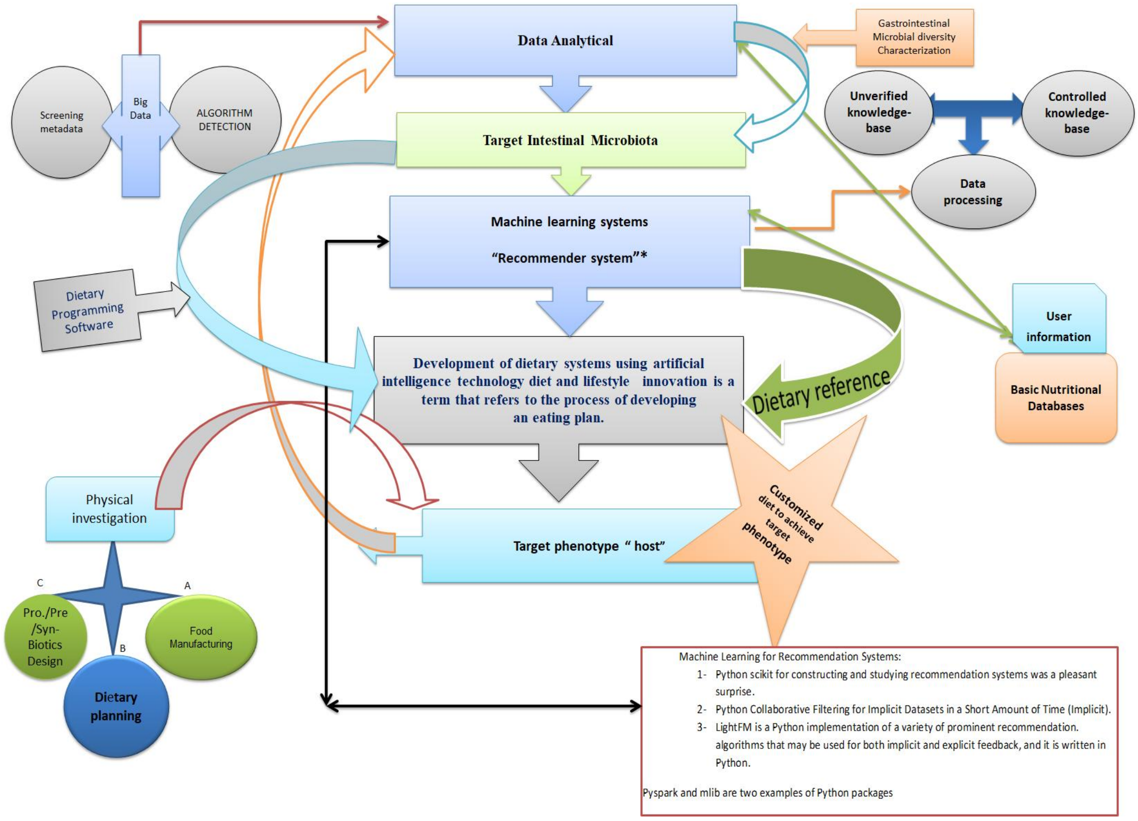

Figure 3 The characteristics of a certain microbiome advised for people to attain their long-term objectives are discovered utilizing information technology approaches

Concepts

You can use tools to help you figure out what you've learned from Microbiome studies. When these bioinformatics techniques look at high-dimensional data like DNA sequence reads [52, 53] and High Spatial resolution Local Mass Spectroscopy (MS) signatures [54], they can look at low-dimensional biological factors like the relative abundance of different types of microbial species and their metabolites. Depending on the quality of the data, the size of the sample, and the study's goal, different information dimensions used, such as gene-level or co-expression metaphysics conceptual frameworks Figure 3 [55].

Computational data processing for the microbiome composition
Meta-omics surveys that would use a lot of data can indeed show really how microbiological communities are used [56-58]. MS technologies can be used to look at this. How much of the gut microbes' revolutionary omics-based strategies content there is can be found out by DNA sequencing [59, 60]. This gives the gut microbes their useful traits. If you want to learn more about your gut microbiome, you usually use a process called "Decoding of a DNA fragment." How many times the hypervariable 16S genes are spliced is used to figure out the taxonomic units that are in use at the time. the fundamental unit of categorization and phylogenetic order, as well as a primary level of natural biodiversity, can help you figure out how many different taxa there are in a microbiome sample by looking at database systems like bioinformatics Tools, Taxonomic databases for the 16S taxonomic categorization includes Greengenes dataset system. http://greengenes.secondgenome.com/. Greengenes is a packed 16s ribosomal RNA knowledge that offers a vetted ontology focused on homologous recombination trees reconstruction and SILVA-ACT - derived from the Spanish silva, which means "woodland" www.arb-silva.de/act "a web-based repository featuring annotated ribosomal RNA coding genomes from prokaryotic and eukaryotic domains. and counting how many of each taxon is present in the sample. The technology called whole-genome metagenomics can show how the microbial community is organized. It can also show how many genes, taxa, common functional groups, or over-represented pathways there are in the community, as well as how many of them there are. Samples with genetic, taxonomic, functional, or metabolic pathway profiles can be used to figure out how diverse a sample's microbiome is both inside and outside of the sample [61]. To measure α diversity, the Shannon index [62], or Shannon Diversity Index (sometimes called the Shannon-Wiener Index) [63, 64], the Chao-1 index (Shannon Equitability Index) [65], and information-rich abundance-based estimators (ACE) [66, 67], are used. To measure β diversity, machine learning distinctive fractional statistic, in data science, it is termed as UniFrac, an online tool accessible at http://bmf.colorado.edu/unifrac, may be used to compare microbial populations in either a linear proportion or quantitative and weighted, or binary weighting, or qualitative and un-weighted representations [68-78], and dissimilarity matrices were made using the Wisconsin polar ordination technique or the Bray–Curtis method or the Sorensen distance ("Sorensen or Bray–Curtis"). Microbiota stability may be measured using the same metrics of diversity, or more complex eigenvalue-based analysis, in empirical investigation. To figure out how many species (i.e., species richness) there are in a sample, jackknifing and bootstrapping are used. The jackknife algorithm is a way of estimating the diversity and distortion of a significant population. It essentially involves an end-up-leaving technique for estimating a parameter (for example, the average) in a data collection of data points (or records). Theoretically, N-1 algorithms are applied to the statistical model, having different factors eliminated from each model [79-82].

A data-driven approach to the development and assessment of the diet questionnaire method
The knowledge gained from intestinal microbiota screenings may be used to determine the diversity of dietary factors. The outcome of this research will be valuable if you are looking for wonderful data such as MS biomarkers and metabolomic data, as well as crude metadata such as western cultural or Middle Eastern dietary patterns from participants in the research. Dietary data are often collected via the use of questionnaires, either through self-reporting or through the use of a certified investigator. An individual's food intake may be described via the use of dietary frequent survey questions or a 24-hour dietary record. Food intake is described either every 24 hours or over a longer period by listing the food items consumed [83-89]. Nutritional records may be utilized in instances when data is captured at the moment of consumption, including when individuals use smart phones to record what they consume Figure 4, such as while they're on the phone [90, 91]. Structure of the food list developed for the app. The food list is based on the Dutch food composition database [92, 93] Nutritional database warehouse characterization includes comprehensive groups of evidence and information on nutritive value relevant aspects of food commodities, including data for macro and micronutrients such as protein molecules, simple carbohydrates, hydrocarbons, essential nutrients, as well as other key food constituents (such as fiber). Dietary Guidelines for Americans Who Want to Eat Well (Dietary Guidelines for Americans) [94], Applications that have been presented, the current state of the art, and future advances are all discussed in detail. Knowledge about various foodstuffs or the abiotic factors for specific compounds is increased as a result of FAO linkages, which include the digital Pharmaceutically Important Ingredients Data Database server, the United States nutraceutical metadata repository https://data.nal.usda.gov/dataset/usda-database-flavonoid-content-selected-foods-release-32-november-2015 [95], and the French Phenol-Explorer online database " http://phenol-explorer.eu/ [96], all of which are incorporated into EuroFIR "European Food Information Resource Network" (www.eurofir.org) [97], and Food Data Central https://fdc.nal.usda.gov/fdc-app.html# as discussed further below. There are many databases, including the computerized Bioactive Substances Information System, the United States flavonoid database directory, and the French Phenol-Explorer online archive, that may be accessed. Internationally recognized rules and procedures that regulate the interconnection of FDBs are the responsibility of InFOODs (International Network of Food Data Systems), the FAO's organization https://www.fao.org/home/en/ [98], which is based in Rome. UN Food and Agriculture Organization Platform of Food Cloud Platforms (United Nations Food and Agriculture Organization), United Nations Food It serve as a platform for worldwide food composition coordination and harmonization by using an international network of provincial data centers in line with the current state of affairs. Together with In Foods, we seek to promote better nutritional requirements throughout the world by bringing together entrepreneurialism, eco-processes, agricultural production, improved health, and diet. Regularly, the platform publishes articles regarding food composition and other food-related topics, and its website provides a downloadable food data warehouse. Standards and harmonization of food composition information from various nations with diverse metadata are required to enable effective data linking and retrieval of relevant information. As a result, tools and processes have been created to ensure interoperability across databases [99]. One such instrument is the language LANGA is an abbreviation for "langua alimentaria" or "language of food," and it is a food description thesaurus that offers a consistent vocabulary for describing foods, especially in the classification of food items for information retrieval. Each of their nearly 40,000 items is defined by numerical qualities that include information on food composition (nutrients and pollutants), food consumption, and food regulation, among others. A link between these dietary qualities (descriptors) and common language phrases in several natural languages is established by Langual [100]. When used in conjunction with many different food data banks from various countries, this important tool facilitates the linking of data to many different food data banks by interpreting different designations and resolving any ambiguities to ensure the correspondence between foods and their attributes, thereby

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

contributing to coherent data exchange [101, 102]. As of now, the Langual food indexing system takes into account factors such as the source of the food (such as the kind of animal or plant species), the preservation of the food (such as freshness or frozenness), cooking, and packaging. It should be noted that the future iteration of this European Foreign Language Thesaurus is considerably more complicated and extensive. FoodOn, a worldwide endeavor now under development, is concerned with highly complete semantics, which includes descriptors for food safety, food security, agricultural methods, culinary, nutritional, and chemical substances, and processes [103]. Experimental scientific breakthroughs are built on data, including data from the food distribution databases, and such figures are often put into models, resulting in outcomes from which conclusions are derived and retracted. Nowadays, these operations may be readily automated by using a bot to download data from food distribution databases [104], which can then be analyzed with the aid of artificial intelligence, allowing for the quick discovery of trends and patterns, among other things. Although the capacity to provide trustworthy and meaningful findings has increased in recent years, the rigor and methodology of the study remain as important, if not more so, than the rigor, detail, semantics, and structure of the database from which the information was derived. Apps that have been specially built may give insights into more obvious links (for example, the relationship between dietary intakes and health) as well as less obvious ones (as between food composition and climate change). Food distribution databases may thus be used for a wide range of purposes other than their conventional use in nutrient intake estimation for diet planning [105]. This is made possible by the availability of IT technologies that make it simpler to handle and analyze vast amounts of data and information. Consequently, food distribution databases (FDBs) can be useful tools for investigating the relationship between foods, diets, and nutrient intake, particularly about nutritional needs and micronutrient deficiencies [106]. However, as state-of-the-art knowledge has revealed an increasing number of compounds from foods that have important physiological roles, there is a growing need to better categorize bioactive compounds in foods. It is just a small number of the more than 26,000 different, defined biochemicals contained in our diet that have yet to be measured, including the major nutritional components found in food distribution databases www.unrisd.org/rpb38 [107]. So, how do we make the move from dietary data to food composition database management systems (also known as FCDBs)? In numerous countries, food composition data (FCD) are collected by a variety of methods, such as those described below.

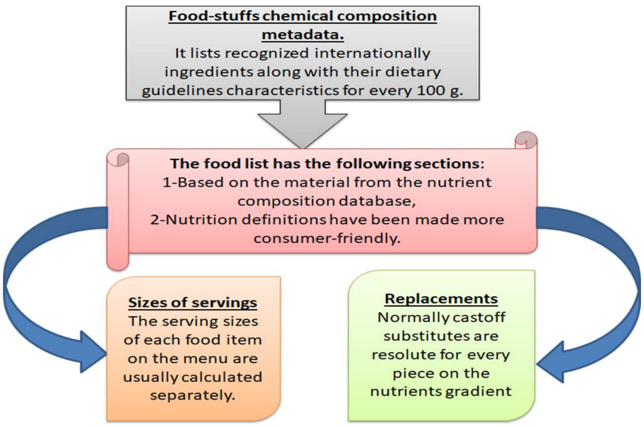

Figure 4 The use of a smart nutrition tracking tool to encourage healthy lifestyles

1 - Chemical composition of food samples may be determined by analysing them (examination using chemical methods). The chemical analysis of samples that are representative of the foods consumed by a particular country's population is the technique that is most often utilised for the creation of FCD. The food samples are chosen with care in line with a specified sampling procedure, and the results are recorded. The food samples are collected, transported to an analytical facility, and stored in a secure area until they are required. In certain cases, further preparation and boiling may be necessary before the analysis, which is carried out under the applicable analytical methods.

2 - Determine how much of each item you plant retained by calculating yield and nutrient retention factors (Formulas). Because of limited resources, it is not feasible to determine the levels of each nutrient present in every food that is eaten around the globe. Therefore, FCDB compilers use several methods to determine the nutritional values that must be included in an FCDB. Consider the following example: to estimate the values for a cooked item, raw food or dish values are often combined with information on projected weight and nutritional changes as a consequence of cooking. This method is shown in the following example. Adjustments for weight change (or yield) during cooking, as well as changes in nutritional content during cooking, are included into this technique of calculating recipe yields and weight changes (e.g. vitamin losses). In addition, the 'borrowing' and/or 'adoption" of nutritional values that were originally estimated by another organisation is a method that FCDB compilers often use. The usage of this mechanism ensures that any borrowed information will be consistent with the compiler's database, which is checked by the compilers.

3 - In this case, values were obtained from one food composition database (FCDB) and applied to another. Data from other countries' FCDBs and information provided by manufacturers are two of the most common data sources used to get borrowed information (e.g. from food labels). Compilers will check the data from any of these sources when constructing their FCDBs to verify that it is of good quality and that the meals included within it are relevant to the database. Comprehensive information on the meal, its nutritional value, and the methodology used to compute it will be provided so that compilers and users of the database may have access to this information as well.

4 - one's measures may be derived from plausible explanations such as existing studies or dietary codes. As stated above, one of the most important goals of the network is to guarantee that high-quality data is accessible in the FCDBs that are linked to EuroFIR. As a result, skilled and experienced technical professionals are always attempting to enhance the content and quality of the data included in their FCDBs, which is an ongoing process. To give the highest level of openness possible, all data must be properly documented before they can be published in the FCDBs and made accessible to all data consumers throughout the world. Aside from that, they must be compiled in line with precise and standardised quality assessment standards, as well as aggregated, validated, and verified again. One of the FCDBs' long-term goals is to provide a quality index for each value, with the ultimate aim of increasing the usefulness of the data for all users by increasing its overall utility.

In terms of nutritional proximate formulation, where can one find data warehouses?

It is often known as Metadata on the physiochemical properties of foodstuffs or nutrients configuration databanks? The program features 17 nutrients configuration databanks with so much more than 35.000 items that have been merged into it, and it also has 17 dietary reference intakes databases that are provided. Moreover, it has one of the most modern Arabic databases on food composition. In the database, there are 2016 food items. 30% of them have benchmark serving sizes visuals, and there are also other ways to figure out how much food you should eat based on the picture. The database and myfood24 have been translated into Arabic for the convenience of Arabic-speaking customers. The GCC's 51 million citizens will benefit from these tools when it comes to evaluating their food intakes. Future development will include more dishes from other Middle Eastern nations to accommodate the nation's over 400 million Arab residents. In the database, there are 2016 food items. 30% of them have benchmark serving sizes visuals, and there are also other ways to figure out how much food you should eat based on the picture (Table

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

1). The database and myfood24 have been translated into Arabic for the convenience of Arabic-speaking customers. The GCC's 51 million citizens will benefit from these tools when it comes to evaluating their food intakes. Future development will include more dishes from other Middle Eastern nations to accommodate the nation's over 400 million Arab residents.

Table 1 The composition of nourishment databases listed below

GITHUB ID	Profitability of datasets	Project website	Reference
Arabic FCDB of GCC foods, with a focus on Saudi Arabia and Kuwait	Yes	https://www.myfood24.org/dataset_	[108]
Egyptian food composition database	Yes	https://globalnutritionreport.org/resources/nutrition-profiles/africa/northern-africa/egypt/	[109]
Spanish food composition tables and databases: Need for a gold standard for healthcare professional (CESNID Foundation, Spain	Yes	https://www.aesan.gob.es/AECOSAN/docs/documentos/nutricion/Plan_Colaboracion_INGLES.pdf_	[110]
Composition tables of foods and Greek dishes (CTFG, HHF)	Yes	http://www.hhf-greece.gr/tables/home.aspx?l=en	[111]
Brazilian Food Composition Table (TACO)	Yes	http://www.tbca.net.br/	[112]
Canadian Nutrient File, Government of Canada (CNF, GC)	Yes	www.foodb.ca .	[113]
Food Composition Database for Epidemiological Studies in Italy (FCDBE, IEO)	Yes	http://www.bda-ieo.it/_	[114]
French food composition (CIQUAL, ANSES)	Yes	https://www.anses.fr/en/content/anses-ciqual-food-composition-table.	[115]
Irish Food Composition Database (IFCD, UCC)	Yes	www.fsai.ie http://www.ucc.ie/en/ifcdb (http://www.eurofir.eu)	[116]
ARFenol-Foods, Contenido de Compuestos Fenólicos en Alimentos Argentinos, Instituto Superior de Investigaciones Biológicas (INSIBIO), CONICET, Argentina, 2018	Yes	http://www.unlu.edu.ar/~argenfoods/Tablas/Tabla.htm	[117]
McCance and Widdowson's The Composition of Foods integrated dataset (Co FID)	Yes	https://www.gov.uk/government/publications/composition-of-foods-integrated-dataset-cofid#history	[118, 119]
Mexican Food Equivalents System (Mexico)	Yes	http://www.dof.gob.mx/nota_detalle.php?codigo=5340693&fecha=15/04/2014.	[120]
Polish Foods Composition Database (PFCT, NFNI)	Yes	https://www.nfz.gov.pl/download/gfx/nfz/pl/defaultstronaopisowa/349/43/1/raport_-_cukier.pdf	[121]
Portuguese Food Composition Table (TCAP)	Yes	http://portfir.insa.pt/	[122]
Swiss food composition database (FSVO)	Yes	SWISS NUTRITION POLICY 2017–2024 http://extwprlegs1.fao.org/docs/pdf/swi175999.pdf	[123]
Tables of the Nutritional Composition of Food Consumed in Brazil of IBGE (TCNA, IBGE)	Yes	https://www.fao.org/nutrition/education/food-dietary-guidelines/regions/brazil/en/	[124]
USDA Food Composition Database, April 2018 (United States)	Yes	https://fdc.nal.usda.gov/download-datasets.html	[125]
Turkish National Food Composition Database	Yes	http://www.turkomp.gov.tr/main	[126]
Ministry of Agriculture of Morocco	Yes	http://ww2.fmp-usmba.ac.ma/.	[127]

Individually designed eating plan databases may be found in Strategic Nutritional Data sources.

Acknowledge approach about how food is made has a direct connection to how the body works. In 2022, Wishart and his colleagues looked at the database healthy human Biomarker Repository (HMDB) (https://hmdb.ca) version 5.0 [128-132], which is based in Canada and is supported by the same groups as Nutrient composition databases vs.1. When they were thinking about metabolic pathways and the benefits of bioactive compounds in humans, they looked at this database. If you want to know more about small-molecule metabolites that the body makes, you can go to a database called A database of knowledge on the metabolic reactions This database is known globally as HMDB, which has a lot of information about them. If you want to study metabolomics and other things like clinical chemistry and biomarker discovery, you can use this tool. This database includes chemical products, diagnostics, and information about how the body works. It also includes more than 100,000 cannabinoid entries (dissolved in water and hydrophobic chemical properties metabolites) that are either abundant (more than 1 M) or rare (less than 1 M) (less than 1 nM) [133]. There are almost 6000 protein sequences linked to the biomarkers above. The whole dataset can be found online. It makes no difference if this directory does not include information on how food is prepared. It is very essential in this context since nutritional research results utilize it to determine whether a diet or keto might alter metabolism, either positively or negatively. If you're looking for anything particular, you may start by searching the Human omics Repository; This online database is well-known throughout the world daily as "Human -Metabolome –Database (https://hmdb.ca), for case studies, codes, chemical compositions, and database recommendations to narrow your search results. People may get individual molecules and transcription factors from the Human Molecular Diversity Database (https://hmdb.ca), which was freely available at the time this article was written. They may also come up with novel search queries based on their molecular mass, elemental composition, or code lookups to find relevant molecules in a given compound. Clinicians may also present their performance on biological information and bioinformatics with machine learning techniques by using MGnify (http://www.ebi.ac.uk/metagenomics), an EMBL-EBI "European Bioinformatics Institute" an online platform that contains the Human Gastrointestinal Protein collection as well as a dataset on the Human Gastrointestinal Genetic Code. There are indeed different understandings of MGnify's progression [134], including the difference in microbial adaptability throughout population groups [135]. Downloadable datasets have been assembled by Sydney University's website Glycemic Control Level Survey Conducted https://glycemicindex.com/about-gi/ [136]. These resources provide you with the glycemic control index of each meal that you look for on their search engine. You may get quantitative information regarding macro and micronutrients, vitamin supplements, and trace elements in gluten-free foods from the Gluten-Free Food Database (Austria) [137]. In particular, concerning being accessible via the research collaboration website Open Science Framework, this database may be edited and added to by anybody who has registered with the platform.

The Food Platform Regarding Industrial Processing (FPIP) is a database that offers information about applications in the food manufacturing business

It could take a long time to discover the precise thermal processing parameters that are necessary for safe food. It is not always easy to determine the "D-value" and "z-value" of decimal reduction time as a function of temperature parameters [138]. They characterize the parameters of the thermal death of microorganisms that are aimed for ingestion in food. The OWL University of Applied Sciences and Arts's Institute for Food Technology NRW is behind the effort. In addition, it includes information on these features that may be employed to create pasteurization or sterilize techniques with a major focus on beverage spoiling bacteria such as E. coli, among others. For example, pH, Brix (symbol °Bx) (the sugar content of an aqueous solution), and water activity (wa), among other parameters, that could impact D and z values [139]: The data is sorted into groups depending on the kind of microbe and medium used, as well as the research from which they were acquired or a cluster of data that was crucial to the experiment. Furthermore, food engineers require the Database of Physical Properties of Food, which may be obtained on the National Engineering Laboratory for Food Engineering website at http://www.nelfood.com (National Engineering Laboratory for Food Engineering) (accessed on August 17, 2021). A project titled the "Physical Characteristics of Food Data Base" was established to collect trustworthy and important information on food's physical attributes and make them accessible online. Working collaboratively alongside NEL, a team headed by Dr. Paul Nesvadba finalized designing and implementing NEL's internet component, which had been sponsored in part by the European Union and to a smaller extent by Nestle, RHMS, and Unilever. More than 11,000 bibliographic references and 1519 resources are included in the database for premium customers, along with 1694 experimental datasets. There are 260 physical properties represented by these files, grouped into 24 food categories, 249 food subcategories, and 24 food subclasses, each of which includes 249 food subclasses. Data may be accessed in the NELFOOD database on five main areas of physical attributes. For instance, the following foods have specific optical properties: (a) food products' mechanical and durability characteristics; (b) natural foods' absorption and mass transfer parameters; (c) food products' electrical and electromagnetic abilities [140].

Food stuff repository that include knowledge on the effect of food on the climate

Farmers have an enormous influence on biodiversity loss, water usage that exceeds replenishment capacity, soil deterioration, and pollution [141], which makes today's food production unsustainable in the majority of cases. The United Nations Secretary-General, António Guterres, was in charge of the 2021 food chain summit, which was managed by the United Nations Food Production Agency, taking Action for Long-Term Sustainable Growth in the Twenty-First Century. The meeting was part of a larger initiative called the Last Century of Action [142]. Those who consume food, on the other hand, are becoming more conscious of the influence that their purchases have on the environment. To achieve this, there must be changes in the way people consume, and food producers must be prepared to make the adjustments that are necessary to accomplish this. There must be a well-defined set of objectives and metrics in place, and these objectives and measures must be supported by a thorough knowledge of the food–environment interaction. While there are still gaps in how data on the environmental impacts of food is compiled, we don't know if the Food and Agriculture Organization (FAO) of the United Nations will continue to make progress in integrating data on the environmental effects of food soon [143]. The environmental footprint of foods is currently included in just a few food databases, and only a handful of these databases are competent for analyzing their environmental effects. One of them is the United States Department of Agriculture's "Experimental Foods" database, which contains information on environmental inputs and outputs on supply chains, among other things [144, 145]. However, this information is not always made available to the general public. A dataset on the environmental repercussions of food on the environment, as seen from the views of producers and consumers, was given by Poore and Nemecek in 2018 [146]. The French Environment and Energy Management Agency "ADEME" has developed the Eco-score® food database in conjunction with the recent launch of the AGRIBALYSE® program (a research program dedicated to the production of life cycle inventories (LCI) for agricultural products). The Eco-score® food database provides an environmental score (Eco-score) for over 2500 food products based on their life cycle analysis (LCA). Despite this, this database has already come under fire, particularly from organic agricultural advocacy organizations, for favoring intensive farming systems while failing to consider the effects of these systems on biodiversity, animal welfare, or the influence of pesticides [147]. This database has also come under fire for favoring intensive farming systems while failing to consider the effects of these systems on biodiversity, animal welfare, or the influence of pesticides. It has already been shown that life cycle analysis, often known as "LCA," on which the AGRIBALYSE® tool is based, is unsuitable for comparing agricultural systems, which has led to criticism of the technique. As a consequence, the development of a better instrument for informing public policy would be necessary [148].

Approaches and opportunities in nutritional database management systems: current and innovative uses

A lot of people are becoming more concerned about how food systems affect people's health and the health of the environment Figure 5. Information that will help us better understand how these different fields work together is very important. It's hard to make progress on all of these fronts because there may not be enough information about what people eat daily. Data makes it easier for people to make laws that work better, and the other way around. The World Health Organization says that more than half of the countries in the world don't keep track of the statistics they need to see if they're making progress toward their dietary goals.It is projected that food staff conformation database will be able to supply this information in response to the growing need for curated and organized information on the composition of food secondary raw materials, novel foods, and/or nutritional sources. This is in reaction to the continuous change of food systems (such as insects and microalgae). Consequently, current concerns regarding the composition of food data may probably become even more acute as a result of these challenges. According to Durazzo et al. 2021 [149], in addition to the inherent features of meals, aspects relating to the extraction and analytical techniques should also be considered. The reason for this is that various extraction processes, as well as varied analysis approaches and methodologies, might result in radically different datasets when compared to the same datasets. This group of researchers also asserts that research has concentrated on a small number of substances within a single class of chemicals and that there are knowledge gaps in the field of adequate analytical procedures for the study of meals. Many linkages are required as a consequence of the commonly acknowledged complexity of food (in its various facets), including information on the relationship between public health and the environment, as well as the relationship between consumer choice and human health, among other things [150-152]. Following up on some previous criticisms, the authors point out that FDBs need to adapt to the always-changing food scene, as well as be refined and standardized to promote international comparisons of study findings. To put it another way, there will be a substantial need shortly for more comprehensive and holistic food databanks (FDBs), which will not only include nutritional content but also other food features. As a result, FDBs will be more effective tools for addressing global health concerns; however, gathering all of the necessary data will necessitate a significant amount of scientific work, especially given the hundreds of new industrially processed foods that are introduced to the market each year around the world. A food composition database such as USDA's or Canada's nutritional file may be queried once the questionnaire survey is concluded to calculate levels of consumption of macro and micronutrients (lipid, carbs, and protein) and micronutrients (vitamin supplements and essential nutrients), as well as dietary metabolites. It's vital to highlight that the microbiota of a person's diet may be investigated using metagenomic sequencing if they aren't previously known (e.g., probiotics with predetermined strains). Some research focuses on conducting direct metabolic characterization of food intake, whereas others rely on metabolic profiles that have previously been defined. A key challenge with any form of study is that food composition databases aren't particularly good. Not even a tenth of the nutritional compounds that are presently known can be described Figure 6.

Multiomics investigation and monitoring may derive from knowledge network-based big data

In certain cases, data on a host may be incredibly high-dimensional [153], for example, high-throughput genome sequences], whereas, in others, it can be quite low-dimensional [154], (for example, obese versus non-obese) depending on the kind of information. In addition to whole-exome sequencing (WES) [155, 156], and genome-wide association studies (GWAS) [157], this information may be complemented by predicting the whole genome sequence for each individual using genotype imputation software, as has been done in several previous studies. To directly analyze the transcriptomic profiles of hosts, microarrays [158], and RNA-Seq [159], may be employed. Alternatively, GWAS data can be used to estimate the transcriptome profiles of hosts using approaches such as PrediXcan [160]. When combined with metabolic pathway databases like MetaCyc https://metacyc.org/ [161], Kyoto Encyclopedia of Genes and Genomes "KEGG Orthology), https://www.genome.jp/kegg/docs/relnote.html [162], Reactome Knowledgebase https://reactome.org [163], or Gene Ontology Consortium (GOC) [164], the genetic and transcriptome profiles may be condensed into relevant, lower-dimensional characteristics that can be used to guide future study and investigation. At the moment, only a tiny number of microbiome researchers are engaged in this kind of research endeavor. Through medical records and surveys, information on other relevant factors such as age, gender, and ethnicity can be gleaned, as can information on body weight, blood pressure, dietary restrictions, and illnesses of a host organism. Information on these factors can also be gleaned through interviews and questionnaires [165].

Computer-Based Analysis (CBA)

EA Eloe-Fadrosh (2022) [166], reported that Multi-omics statistical data on microbiota composition may be investigated and obtained through the national public Microbial Community Data Group effort (NPMCD) public data platform (https://data.microbiomedata.org). This is accomplished effectively by the use of a linked, distributed information structure that adheres to the FAIR data principles (Findable, Accessible, Interoperable, and Reusable) . At the moment, the NPMCD Data Portal contains 10.2 terabytes of multi-omics microbiome data from two DOE user facilities: the Joint Genome Institute (JGI) at Lawrence Berkeley National Laboratory (LBNL) and the Environmental Molecular Systems Laboratory (EMSL) at Pacific Northwest National Laboratory (PNNL). This data includes metagenomes, metatranscriptomes, metaproteomes, metabolomes, and natural organic matter characterizations. The supervision and assimilation of statistics are simplified by a highly adjustable data diagram (https://github.com/microbiomedata/nmdc-schema) built on community-driven standards. Environmental context, instrumentation, and functional characteristics may all be searched for in annotated multi-omic data products created by the NPMCD methods and linked through shared biosamples. The NPMCD Data Portal is a prototype system that includes download capabilities and numerous search components, such as an interactive geographic visualization of samples; an interactive Sankey diagram that depicts the distribution of environmental categorization; a time-series slider for selecting longitudinal samples of interest; and an upset plot that depicts the amount of multi-omics data collected from the same bio-sample within a research project. However, the computation and assessment of Microbiome data has been the subject of several reviews. In this section, we will discuss big data and analytics, algorithms, and artificial intelligence-based recommendation systems that may be used to create Microbiome-aware technologies that are concerned with nutrition and wellbeing [167-170]. People who study these things want to know how things work and what could go wrong. They also need design plans of real-world situations they can use in their own research and development. Specific examples and illustrations of the machine learning algorithms covered here would be found at the following citation [171-173].

Data mining tools for the assessment of Microbiome knowledge

Targeted amplicon sequencing of the Microbiome may provide information on the diversity and activities of microbial communities. Due to rapid advancements in sequencing technology, researchers have been able to examine microbial communities to an unprecedented depth. Amplicon sequences used to identify taxonomic groups. Alpha and beta-diversity, as well as their relative abundances, are calculated using data from taxonomic analysis. Sequence quality screening and alignment, phylogeny creation, and sequence alignment are all necessary computer techniques for these sorts of studies. Several publicly accessible microbial community knowledge processing methods and pipelines are downloadable that can create the different types of data addressed in the literature. Several accessible to the public Microbiome data processing techniques and pipelines are available that can create the different forms of data addressed in this paper. The following table shows a sample overview of such techniques and pipelines: Table 2: Methods and Pipelines it comes to 16S data, QIIME www.qiime.org and MOTHUR https://mothur.org/ create a bigger array of possibilities for the user when compared to UPARSE, although all three algorithms are successful. QIIME 2 https://view.qiime2.org is quickly establishing itself as alternative to its ancestors, in part owing to its flexibility and level of customer support. Techniques like Kraken , MEGAN, MetaPhlAn2, and HUMAnN are utilized for metagenomic sequencing, as are other techniques. Possibilities and Obstacles in Data Processing for the Microbiome: As the number and complexity of data processing tools increase, researchers and practitioners face both opportunities and challenges. Although recommended practices have been proposed, tools remain far from being completely automated and leading to accurate outcomes. Additionally, since the encoding coverage of genomic sequencing assemblies is less uniform than that of single genome assemblies, the assembly of metagenomic datasets is

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

more difficult. Because there are so many uncharacterized microorganisms (known as microbes dark matter), alignment and misalignment issues are magnified by this. There are many ways to evaluate the technique and conclusions of various researches, and each study may use a different approach or a different application of the same method. Due to variances in library preparation (e.g., primers used), sequencing machines and their settings and non-uniform ways of structuring and storing for microbiome data, data collection, and integration of microbiome data from diverse research are problematic due of several variables [174-179].

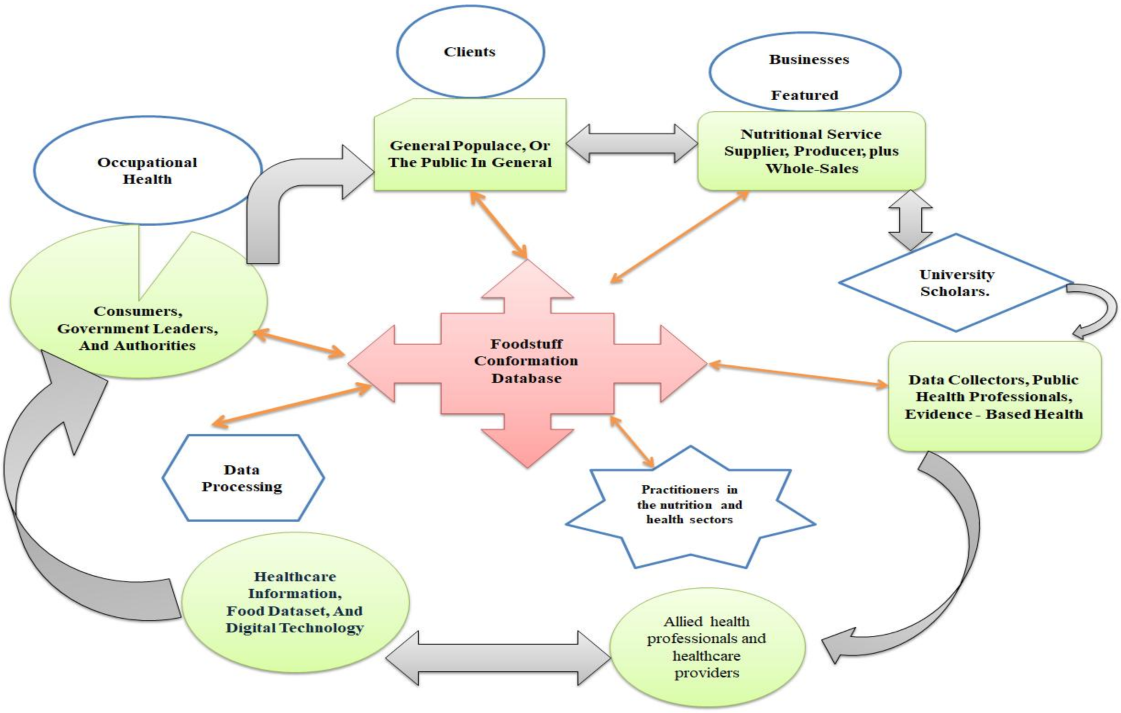

Figure 5 Diagram depicting the many application areas of dietary guidelines metadata

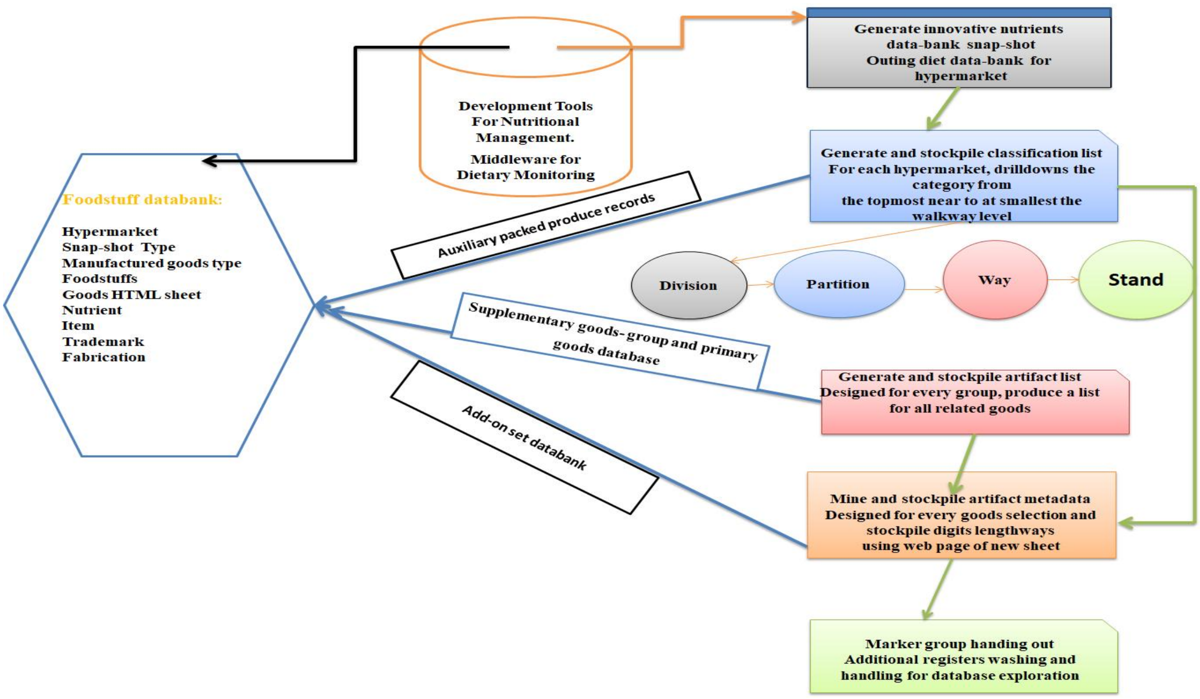

Figure 6 The screenshot dataset compilation mode in the nutrition data system

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

Table 2 Intestinal microbiota data processing issues and innovative solutions, mostly with illustrations and viewpoints on how to improve them

Phases	In this example, we're going to look at some of the steps	Emphasis on strategies and effective usage in common processes	Reference
Acceptance testing	Deletion of mutants and minimization of interference	Trimmomatic is a computer application that would be very fast and multiprocessing. It could be used to start cutting and snipped Sequencing was performed (FASTQ) data, and also to extract dongles from the data sources. (http://www.usadellab.org/cms/?page=trimmomatic)	[174-179]
	Eliminate any measurements that have been contaminated by the caller's Mtdna.	Amplicon Noise is a set of tools for background subtraction from 454 sequenced PCR amplification products. It consists of two steps: attempting to remove interference from of the decoding stage and stripping away PCR juncture inconsistencies. The above venture entails the Three - headed optimization technique for cloned discharge.	[176]
		UCHIME is a novel algorithm that identifies chimeric sequences composed of two or more segments. A database of chimerism-free sequences is either used or new chimerisms are found out of the blue by looking at how many of each type there are. http://drive5.com/uchime.	[177]
		Deblur picture in a single file Blurring artefacts are removed from photographs using the technique of deblurring. Unsharp masking (also known as deblurring) is the process of recovering a sharp picture S from a blurred image B. S is highly integrated using K (the fuzz core) to generate B. https://github.com/biocore/deblur	[178]
		The Dada2 pipeline is comprised of two components. Dada1 refers to the first portion of the equation, while Dada2 refers to the second part of the equation. To put it another way, dada2 expects the inclusion of a distinct fastq file for each sample (or two fastq files, one forward and one reverse, for each sample). Whenever possible, demultiplexing should be performed at the sequencing centre; however, if this has not been done, a number of tools are available to assist with this task, including the popular QIIME python scripts split libraries: fastq.py followed by split sequence file on sample ids.py, as well as the programme idemp, among others. If demultiplexing is not performed at the sequencing centre, it should be done at the sequencing center. https://benjjneb.github.io/dada2/	[179]
Operational Taxonomic Unit (OTU)	Readings are assigned to OTUs based on their names.	It is generally agreed that the operational taxonomic unit, often known as OTU, is the fundamental unit of numerical taxonomy. Each one of these units could be used to refer to a single person, a group, or a type of thing. It is common for microbial ecology to use OTUs or agglomerations that have complementary sequences at the molecular scale as specific indicators. https://motu-tool.org/	[180]
		UPARSE-OTU is a technique for producing clusters (OTUs) using next-generation sequence alignment reads of genetic variants like 16S rRNA, the fungal ITS region, and the COI genome. The statistically representative sequences of OTUs are more reliable predictors of genomic sequences than genomic sequences, and the quantity of OTUs is considerably closer to the number of species than the number of species in the genetic code. https://drive5.com/uparse/	[181]
		Kraken is a powerful tool for labelling metagenomic DNA sequences with taxonomic labels that is both incredibly quick and extremely accurate. Megablast and MetaPhlAn are both very good tools for figuring out how many base pairs there are in a sample. Megablast can do this faster than Kraken in its fastest mode, but MetaPhlAn can do it 11 times faster in its fastest mode. http://ccb.jhu.edu/software/kraken/	[182]
Assessment on the basis of physic--chemical characteristics	characterizing and anticipating the activities	MEGAN6: A full set of tools for looking at microbiome statistical information in an engaging environment. All of the online tools you'll desire are in a native platform. The NCBI framework or a specialised taxonomy, like SILVA, can be used to look at taxonomic relationships. Analytical methods that look at how the body works include InterPro2GO, SEED, eggNOG, and KEGG.	[184]
		[MOCAT2] This is a computer network that is used to build metagenomic sequencing and identify specific genes. It has unique features for phylogenetic and fully working enrichment monitoring, as well as a unique system to predict and build genes. Metagenomes can be quickly and completely assessed for their function through the development of an algorithm and the efficient identification of non-redundant standard catalogues. This is done by disseminating assignments from 18 repositories that cover a wide range of functional categories.	[185]

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

An algorithmic technique to the visualization of massive amounts of data

To turn raw data, like metagenomics sequence reads, into biologically meaningful representations, like Functioning Taxonomic Divisions, that can be used to make sense of the data, data processing is a big part of the process, On a methodological level, it is well-established (OUT) [180]. After all of the processed sample data from different sources has been combined, data analytics can start (i.e., nutrition, gut microbiota biodiversity, and the human GIT) [181-185]. Most of the time, all samples come from the same research, which ensures that the experimental conditions and data processing procedures used are the same across all samples. Due to the limited resources, the researchers must also limit the amount of information they collect to have enough statistical power to test their hypotheses. The recent and huge rise in the number of microbiome studies that use public data has made it possible to combine data from many different studies [186-192]. In these cases, Statisticians who acquire and store data, as well as those who prepare, sequence, and process it, must take great care to avoid making any errors.

Dilemmas in analyzing microbial community datasets

As shown in Table 3, statistical data on the intestinal microbiota might be difficult to interpret. It illustrates some of the issues that emerge while analysing microbiome data, such as how to deal with them. Because the first dilemma is that there are a lot of different kinds of datasets in the microbial population, It's considered compositional when a quantity like the number of readings attributed to a species can only be evaluated in terms of proportions. This information can be easily compared across other samples. Due to technical restrictions, the amount of actual reads attributed to different data characteristics (e.g., OTUs, gene products, and taxonomic groupings) does not reflect integer diversity and abundance, and so judgments should not have been generated from them [193]. As a result, instead, the range of amplicons should be modified to population dynamics and looked at with that intellect. Some randomized trials use rarefaction to make up for variations in academic library size caused by incomplete genomic sequencing sampling. If you want to study the Microbiome through metagenomics, there are a lot of pipeline projects that can do this. However, they are ineffective because they don't keep many reads, which makes them less sensitive at detecting differences in abundance and makes it more difficult to figure out how diverse the Microbiome is [194]. This is because metagenomic data has a lot of different kinds of things that make it hard to understand. Datasets that have a lot of features but a small number of objects are titled "top linear" or "computationally intensive." It takes a small group of people to figure out how many things they have to do with their host body, diet, and Microbiome. This leads to very big datasets with a lot of different things [195]. Unsupervised dimensionality reduction (DR) is a technique that may be used to reduce the number of OTUs in a dataset when there are a disproportionate number of them (explained below). In the third problem, which referred to as an experimental research analysis, it is necessary to look at a large number of distinct hypotheses. In this situation, it has to do with the likelihood that someone will make a mistake when they learn something new for the first time. Several approaches may be used to do this, including expanding the number of individuals who participate in the research and altering the p-value (which will be addressed briefly in the following table). It is because they all began out in the same area that they have developed a complex system of organizational structures over time. As a result, phylogeny-aware strategies must be employed to cope with the issue, and it is possible that assumptions such as sample independence are not correct [196]. It is necessary to remove particular numbers from the data that has been gathered to solve the sixth difficulty, which is addressed below. In the case of marker gene sequencing, it is not possible to determine how many functional genes have been introduced into the Microbiome as a consequence of the process (i.e., how many are misplaced). Unless we have a complete understanding of what each microbial species is doing, it will be hard to research the gut microbiota effectively. Metatranscriptomics data may theoretically be used to estimate this information, but in practice, this information is seldom available. In this regard, PICRUSt https://github.com/picrust/picrust and other gene attribution approaches are useful tools [197]. We may be able to alleviate some of the symptoms of this illness if we use annotated databases for gene restoration in the future. Methods for associating microbiome properties with certain host phenotypes may be classified into two major groups according to whether the desired qualities are identified by reinforcement learning or semi-supervised learning. When using supervised learning, the qualities of interest are identified via observation of the features of interest. When there are no labels on the records being inspected, it is acceptable to use a naive Bayes classifier; but, when there are labels on the records being studied, it is important to use supervised learning. However, while the methods utilized in this instance are more advanced, such as semi-supervised learning, they are still used [198]. This publication does not cover the topics of supervised learning, which makes use of both annotated and unprocessed data, or learning algorithms, which is the transmission of knowledge from one activity to another once it is mastered [199].

Table 3 Limitations and recommendations that inevitably occur during the processing of microbiome data are highlighted here, as well as illustrations and practical implications

Interpreting intestinal microbiota metadata has its limitations	Layout case	Ideas clauses
Variables in constituents include: Omic information processing produces scan statistics for newly identified elements such as DNA, taxa, and source repositories from a specific population, which is referred to as "interpretation counting" in genomics. These started reading frequencies may be significant only within a study population.	Respondent A has eight reads mapped to bacterial lactic acid in his or her feces, while someone else appears to have sixteen reads. Is it reasonable to assume that this bacteria is more prevalent in the gut of user B than in the gut of adult A? Reading frequencies within data really aren't comparable, and this is the case with this method.	Prior to compare read counts, it is required to convert them to proportionate abundances (RA) (RA). If a job is specified in terms of read counts, place a restriction on the total number of readings per sample in the task description.

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

Table 3 Limitations and recommendations that inevitably occur during the processing of microbiome data are highlighted here, as well as illustrations and practical implications (*Continued*)

Interpreting Intestinal microbiota Metadata Has Its Limitations	Layout case	Ideas clauses
The degree of computational complexity is really high: When metagenomic data is processed, various new entities are discovered for each sample, such as genes and species. These entities may or may not be shared across multiple samples as a result of the data processing process. Because each entity is allocated one dimension during the data aggregation process, the number of dimensions assigned to each entity is disproportionately great when compared to the number of sampled entities.	Bioinformatics and metagenomic data Each of the 20 faeces samples tested included the relative abundance of ten different microbial families, according to the results. Classical linear regression may be used to determine an individual's age using relative abundances gleaned from aggregated data. Whenever twenty observations are matched, a total of more than twenty microbial families are revealed.	Prior to developing a linear regression model, do a major component analysis to eliminate as much multiplicity as feasible. A novel technique for predicting the study's result is to use a Normalized Fuzzy Multiple Linear Regression analysis based on the Dynamic Lasso Algorithm. Higher-level taxonomic ranks like phylums should be used to measure how many different types of microbes there are in a given area, rather than just how many different types there are in each type of group.
Some theories are as follows: In addition to evaluating the degree of dimension, high - throughput sequencing data enables the researcher to create a sequence of hypotheses, which ultimately leads to the discovery of designs that are not the result of randomness or randomness in and of itself. In certain circles, this is termed to as "the strong statistical likelihood of moderate likelihood occurrences."	Check out the following image for a visual example: Based on faecal samples from more than 200 people, 50 percent of whom have Ulcerative colitis and 50 percent who are healthy (using 16s rdna algorithms), research was done to figure out how many different kinds of bacteria there are and how many different kinds of bacteria there are at the species identification level using metagenomic data processing. In this study, it found that people who are sick and people who are healthy have 40 different species more or less often. Tests were used to show this (p-value 0.05). Could this be, in practice, the ultimate choice? The traditional p-value threshold is good for one hypothesis, but when there are more than one, the t-test is done 1,000 times, which leads to a lot of wrong findings.	In order to keep the rate of false positives and false negatives as low as possible, measure the false and misleading Exploration percentage rate p-value (also widely recognised as just the q-value).This will help you figure out which Exploration percentage rate p-values aren't true or not true (also widely recognised as just the q-value).
Several relationships are found in a structure that is hierarchical. As a result of genetic and phenotypic similarities, taxonomic variables (like species and OTUs) have known hierarchical links. When it comes to microbiome data, this does not hold true. However, when it comes to genomic data, the assumption of independence doesn't hold true because taxonomic variables (like species and OTUs) have known hierarchical links because they have similar genetic and phenotypic traits and traits. As a result, typical statistical methods that are based on the idea that two variables are independent are not useful in this case.	When comparing two sets of microbiome samples, it is possible to use beta-diversity to evaluate how similar they are to one another. Alternatively, are there any simple methods for determining the beta-diversity of a taxonomic order that can be used in conjunction with computing the usual Euclidean distance between relative abundances at a given taxonomic level? In response to your inquiry, the Euclidean distance does not take into account how similar two species are to one another in terms of physical characteristics.	Phylogeny-aware measures, such as the distinctive percentage of metres or the UniFrac distance, may have been used to replace this technique. These measures, which take phylogenetic relationships into account when computing relationships between two different points in time, are becoming more prominent.
Meta-transcriptomics and meta-proteomics are two methods that have been used to fill in the gaps in metagenomic data. These methods have been used to fill in the gaps in the data. Because of this, microbiome research is very important because it needs to know more about the role of microbiota.	For example, in one instance, metagenomic data processing from marker-gene data has provided us with relative abundances at the genus level, but we do not know the potential activities of the microbiota in terms of the proteins that it may produce as a consequence of this processing. Should we abandon our efforts to do further research? If we don't have personal knowledge of proteins, we can make informed predictions about them, which is why we don't need to	Methodology: Databases like as Greengenes, which include the whole genome sequences of known species from a number of taxonomic groupings, may be used to predict the activities of genes and proteins based on their sequence data.

Instructional strategies with strict supervision

Validating hypothesis testing and comparing results. When you look at how things change, you can use one or more variables to do it. It's possible to use nonparametric tests like Wilcoxon rank-sum or Kruskal–Wallis to test a single-variable hypothesis with these types of tests. This may be accomplished via the use of the student's t-test. Attention deficit hyperactivity disorder sufferers had less variety in their gut microbiota than the general population, according to T-tests [200]. With non-parametric tests, this isn't the case since they don't rely on assumptions about how the data in the sample is distributed. When analyzing data on anticipated routes, for example, a multiple linear regression analysis known as the Frank Wilcoxon rank-sum test "for statistical significance" is employed to analyze the information. According to the above study, there are more metabolites in this procedure called "N-linked glycosaminoglycan Process Followed and Deterioration" in the warmer months than in the cold season. This effectively means that more metabolites are used in the framework [201]. When a statistical test is performed with alternative assumptions, it is critical to modify the statistical significance of the results (i.e., multiple hypothesis testing [202]. Numerical methodologies that regulate the false positive and false negative rate correction can also be used to accomplish this goal (i.e., q-value) or The Holm-Bonferroni Method (sometimes called Holm's Sequential Bonferroni Procedure) is a statistical method for comparing two or more sets of variables [203]. In addition to assessing research questions with a large set of variables, you can use multinomial logistic regression and nonparametric chi-squared test (MANCOVA) [204], or non-parametric tools such as permutational multiple regression assessment of variability (PERMANOVA) [205], or interpretation of resemblances (ANOSIM) to achieve the research objectives [206]. Subsequently, data are divided into the following separate communities, which are then combined into a single large group. Following that, they are amalgamated into a single big bunch (e.g., based on some feature values). When it comes to a certain characteristic, such as a number, you want to understand how much this grouping can explain how values are spread out (response variable). The most often used approach for this is the analysis of variance (ANOVA), which evaluates one response variable with a normal distribution, which is the easiest condition. A recent study, in 2022 by Prochazkova M [207], for example, revealed two bacterial phyla (Bacteroidetes and Firmicutes) in the microbiota of children living in a rural African village and kids residing in Euro, with the wide availability of the bacterial isolates phyla diverging. ANOVA may be extended in the form of multivariate analysis of variance (MANOVA), which is capable of dealing with a large number of answers and is thus more practical. A similar test is commonly used to determine whether the microbiota of children with PRAH-dur VIL-e (PWS) syndrome coincides with autistic continuum abnormality (ASD) and whether or not the microbiota of children with normal body weight changes before and after treatment, according to standard practice [208]. Non-parametric procedures are used in lieu of standard statistical methods since normal distributions do not always hold in all cases. Persons with Crohn's disease were compared to healthy people in one study using PerMANOVA to look for taxonomic differences in the microbiome [209].

A parametric and non-parametric assessment using multinomial logistic regression and correlation

It is possible to determine the degree of relationship between two variables via the use of a correlation test. There are many different sorts of relationships that may be discovered via the use of correlation measures. This is termed the Bray Curtis dataset of heterogeneity [210]. A linear link is quantified by the Pearson correlation coefficient r, whereas the nonparametric version of the Pearson product-moment correlation is quantified by abundance similarities, among other things, according to Bray–Curtis. Spearman's correlation coefficient (rs), often known as the rank correlation coefficient, is a quantitative measure of the strength of relationships between two ranks [211, 212]. The researchers were doing a Simulink comparative assessment of a varied array of correlation measures for microbiome data to determine which one was the most effective. Because they are meant to look at complicated interactions in compositional microbiome data measures such as SparCC [213], and Investigation of town's resemblance Parameter " LSA Local similarity analysis" [214] ,perform better than other metrics because they are designed to look at these relationships. It is also used to examine the TwinUK dataset https://twinsuk.ac.uk [215], If you want to find bacteria that live in your body because of your genes, you can do this. To do this, intraclass correlation data was used to build a network of microbial families. PhILR – Bioconductor has been in use since its widespread adoption in 2021 by SM Devlin et al. [216], PhILR – Bioconductor is an acronym that stands for Phylogenetic Isometric Log-Transformation. If you want to move compositional data into noncompositional space, you can use the Ratio Transform to do that. You can then use the same data analysis methods that you usually use. It's usually best to apply these changes to traits that are both compositional and phylogenetic, but it's not always best to do that. A goal when using regression methods is to predict how a single continuous variable will change by using data from other continuous variables to predict how it will change. One way of thinking about correlation analysis is to think of it as a special case of regression analysis when there is only one variable. There are some applications where standard linear regression can be used to get good results for nutritional human gut microbiota prediction. Variables should be kept in mind when they are linked to OUT abundances, which means that standard assumptions about linear relationships, normal distribution, and dependence no longer hold. There are many ways to predict the composition of OTUs [binarized for adding up to one]. One example is zero-inflated continuous distributions [217]. When attempting to predict the composition of OTUs, this is the desired outcome. It is occasionally necessary to utilize two different regression models, with one half being a logistical model that informs you how probable it is that the given OTU would be present, and the other half being another logistical model. In Part II, we use extensive linear regression using a beta distribution to determine the number of OTUs present in the sample. This is because an OTU has been detected in the datasets [218-220]. The normal practice in the context of phylogenetic hierarchies is to employ closely related species that have evolved considerably, such as evolutionary relationships modified least-square, to account for dependency between data points. The chances of making a mistake in multiple regressions are higher if the evolutionary history of microbial communities is not taken into account. Even though phylogenetic comparison approaches are not often utilized in microbiome investigations at present, this is probably a contributing factor to the high number of false positives that may be reduced if these techniques were applied [221]. If you want to find out how two things (like the abundance of microbiome OTUs and the number of metabolites) are linked together, you can use CCA (Canonical correlation analysis) [222]. Linear transformation pairs that are the most correlated when applied to data are found by CCA. At the same time, CCA makes sure that different linear transformation pairs stay orthogonal. The original CCA, on the other hand, doesn't work when it comes to high-dimensional microbiome data because the number of variables is too high for the number of samples. It may be possible to use regularization to solve this, which could lead to sparse CCA algorithms being made. As part of a study looking at how the gut microbiota and metabolome traits are linked in patients with Alzheimer's disease (AD), a sparse CCA is used [223].

Supervised classification of machine learning algorithms

According to Li et al 2022 [224], the goal of supervised classification is to develop a prediction model (a classifier) using labeled training data. In contrast to regression, where labels are continuous and numerical, classification labels may have binary or categorical values. A classifier was developed in one investigation to infer the distinctiveness of specimen donors based on relative OTU abundance and diversity estimated from 16s ribosomal RNA bowel specimens. Random forest Supervised Machine Learning algorithms were used to do this, which is a technique that creates an ensemble of decision trees [225, 226] If you were in a different study, the goal was to figure out which people were healthy and which people were sick based on relative OTU abundance data (including species level) from shotgun metagenomics sequencing of the gut and other parts of the body described by S Tkach 2022 [227]. Some of the other tools they used were random decision forests and the support-vector machine classifier, which is a computer algorithm that can be used to classify and predict data. The support-vector machine is a powerful method for building generalizable and interpretable models that is well-understood in math [228]. Except for a few datasets; random decision trees classifiers outperformed support vector machine classifiers in their analysis. Random decision forests and support vector machines both offer built-in capabilities for dealing with overfitting concerns that develop in large, high-dimensional datasets. Random decision forests algorithm does this by the use of an ensemble-based method, in which the prediction is formed based on predictions from a large number of trained classification models. In a Supervised learning support-vector machine Classifier, the parameters of the prediction model are limited depending on criteria that have been established in advance. It is important to note that restricting the model parameters is often similar to regularization in terms of mathematics [229, 230]. There are loss functions that are used to figure out how much error there is in the model's predictions as a whole. As in both cases, the goal is to get as low of a value as you can. Because if you use regularisation to cut down on your mistakes, it doesn't just matter how many mistakes you make if you do this. Taking these things into account, the loss function also takes them into account. A regression model with L1 and L2 regularisation is shown in this way, There are two types of regularisation: L1 and L2. In L1, model parameters are scaled and added to the loss function; in L2, the opposite happens. When a large dataset has a large number of features, L1, dubbed Lasso Regression (which produces output in binary weights from 0 to 1), is used to reduce the number. Ridge Regression (L.2) is a method for dispersing error terms in all weights, which leads to more accurate tailored models. If two models both have the same error, a training method will choose the model with the lower parameter values if they both have the same error. Many people use L1 regularisation when picking features. This means only picking features that aren't zero in the model that was used to train it This is because a model with a low prediction error can be made by only using a small number of the features that are available to make the model. An artificial neural network is better at classifying things than other techniques in many different fields, like biology [231-235]. To mention a few examples, machine learning and artificial intelligence are two fields of study. Recently, microbiome data has been utilized to develop a novel artificial neural network (ANN)-based tool known as Regularization of Learning Networks, which is based on artificial neural networks (RLN). We've put it to use to see how well it performs in practice. A reasonable approach to thinking about changing the regularisation parameters of a neural network is to conceive of each parameter as a distinct regularisation coefficient. An effective method of doing this is through the use of the RLN. When each parameter is viewed as a distinct regularisation coefficient, the RLN is a useful tool for adjusting the regularisation parameters of a network's regularisation [236]. Individuals with this kind of BMI and cholesterol may use RLN to figure out their numbers, and they are capable of doing so. So they can predict, for example, how many distinct types of creatures will exist in the future and how many genes each of them will have. Experts believe that new classification algorithms will be developed to cope with the challenges that nutritional human gut microbiome data is experiencing. They will consider a variety of aspects, including biological phenomena, characteristics of measurement equipment, and upstream data processing pipelines, among others [237].

Unsupervised learning for visualization techniques and wavelet analysis

Depending on how many characteristics you have, it may be beneficial to limit the number of features using an unsupervised phase before moving on to a supervised step. An important feature of many unsupervised learning techniques is the implementation of a transform method, which may be utilized to minimize dimensionality. By using high-dimensional datasets, there are a lot of different ways to get a high-resolution and detailed look at things like the gut microbiota. To make predictions in big data analysis more accurate, there may be a lot more data available that can help with that. A lot of data analytics methods, on the other hand, don't work well when they have to deal with huge amounts of high-dimensional data. This means that disaster recovery plans have to be put in place Wavelet Transform. When the number of dimensions in the data is cut down, visual analytics and processing become easier to do. It is one of the most common ways to mine data, It is based on Haar wavelet analysis. a Haar wavelet analysis is used to recreate the original features in the data set. They are a small group of uncorrelated properties that are used. The Haar wavelet transform of a data set is a linear combination of its original attributes. They are responsible for keeping most of the variance in the data set in check, though. As of recently, there have been a lot of studies that say this. In recent research [238], gene abundance predictions from the gut microbiota of human donors were analyzed using Haar wavelet transform. Taking data from a dimensionality of ten million to two dimensions only allowed for the presentation of data on a typical two-dimensional scatter-plot (i.e., Haar wavelet plot) [239, 240], It was discovered that there was a significant variation in the microbiota of Beijing and Scandinavian volunteers. Another study developed a reliable indicator of antibiotic resistance by using the top five discrete wavelet transforms of an individual bacteria's genome that was sequenced from newborn fecal samples [241-245]. If you look at how traits are linked together, you can make decisions based on how they work together. People have come up with many different ways to do factor analysis. We'll go over them in more detail below. Data reduction in multivariable logistic regression analysis takes into account groupings of features to make sure there is an equal amount of each type of character in the wavelet transform. It is an extension of the wavelet transform called multivariable logistic regression analysis. It takes into account groupings of features to make sure that all types of characteristics are evenly distributed in the wavelet transform that is made [246]. The Multiple Factor Analysis method was utilized in one research to visualize the host and microbiome traits in two dimensions at the same time [247- 249]. Experiment with exploratory factor analysis to identify previously undiscovered latent characteristics, also known as factors, that may be utilized to establish the link between apparent data and latent components [250]. In contrast to the factors revealed using Haar wavelet analysis, which is linked together, the components discovered through exploratory factor analysis are isolated from one another. In this case, Haar wavelet analysis is employed to explain total variance rather than correlations, and correlations are utilized to explain total variance. The exploratory research design was used in a recent study to develop four main components representing connections among 25 prominent genera to evaluate the relationship between early childhood neuropsychiatric effects and the microbiome among 309 children [251, 252]. If a structural equation model of wavelet coefficients and their interactions with observable variables is not supported by data, two approaches such as confirmatory factor analysis and multiple linear regression analysis might be applied [253]. Wang S.,2022 conducted a study in which they developed and evaluated a conceptual model to mimic the influence of mother and newborn characteristics on the microbiota of breast milk [254]. A key variable analysis is frequently performed using the Logistic regression model "FactoMineR" specialized to multivariate exploratory research, as well as the R modules [255], and IBM SPSS database management systems [256]. Another way to look at beta diversity is called principal coordinate analysis (PCoA), which is also called multidimensional scaling (MDS) [257], and it can be used to show 2- and 3-dimensional representations of beta diversity in 2- and 3-dimensional ways. As a result, it can be used in situations where the distance between feature vectors from samples isn't long enough (e.g., owing to the noteworthy fraction of zero elements density and phylogenetic tree associations). To compute distances between samples (for example, the UniFrac distance between OTU abundances of a pair of sample donors), PCoA takes as input a matrix of distances between samples (for example, it then assigns new coordinates to each sample, such as PC1 and PC2, in such a manner that the Euclidean distances in the new coordinates are comparable to the distances in the original coordinates in the matrix). PCoA was, for example, employed in the context of UniFrac distances between OTU abundances (derived from 16S rRNA samples) in the gastrointestinal microbiota of donors, which were calculated using the UniFrac distances technique . People who lived in the United States had different bowel microbial communities than people who lived Live outside the US communities [258], two-dimensional imaging showed analysis called linear discriminant analysis (LDA) does the same thing, but it is supervised and closely linked to regression and ANOVA, which is why it is called a minimize dimensionality method [259]. When you use class labels, it is very different from PCA and PCoA. This approach creates new features that are linear combinations of the original features while simultaneously separating samples based on their class labels throughout the process In one case, LDA was used to differentiate between gut microbiota samples based on diet, but not between samples based on minimizing dimensionality [260]. Because of computational complexity, regularisation to accommodate this would be necessary for effective LDA use in high-dimensional sampling, as has been demonstrated for high-dimensional gene expression profiles [261]. The ideal degree of dimensionality reduction (e.g., the degree of main components) varies depending on the data and the job to be completed upstream. When it comes to data visualization jobs, they are generally limited by the constraints of human visual perception (three-dimensional). Most of the time, when we're working with downstream supervised learning; we want to make sure we can get as much DR as possible without having a big impact on our predictions. A good example of this is GP Way et al. [262], who investigated the influence of the quantity of DR on classification performance for gene expression data.

Data mining and artificial intelligence deploy clustering algorithms focusing on densities

If there are similar microbial communities, they should have the same effects on the host. It is important to choose a similarity measure first. When that's done, different cluster analysis methods can be used to look for groups of samples with similar microbiota, which can then be grouped. Cluster analysis of 16s rRNA data from stool samples was used in a recent study by GS Slanzon 2022 [263], to identify three types of microbiota, or enterotypes, that are unique to each person. There were more examples of how this clustering isn't just about the data, but also about what you choose to do when you do it. A study by JN Martinez-Medina 2021 [264]. says that. Cluster analysis results may be affected by four important decisions, which are explained in this article (other than how the data is processed before cluster analysis) (other than upstream data processing). The distance measurement is the first thing to look at. Simple, well-known clustering methods like the Euclidean and Manhattan distances are given by a vast variety of clustering libraries. It is because of previous compositionality-aware transformations like ILR that these metrics can be used. Microbiome analysis is made feasible by beta-diversity metrics such as weighted and unweighted UniFrac distances, which are meant to account for the compositionality and phylogenetic connections of Microbiome data. To better appreciate the clustering results, researchers should pay special attention to the characteristics of the distance measure that was utilized. The clustering method is in second place. In algorithms such as Partition Around Medoids [265], and Hierarchical Clustering [266], various distance metrics may be used. Others, such as k-means, are connected to a single distance metric but are less computationally intensive than the first two methods. The next factor to consider is the fuzzy set. When using clustering methods, it is common for the clustering algorithm to be supplied as an input. When a number isn't known, the highest cluster score is used to determine the winner. Several prominent cluster scoring measures, including projection quality [267], pattern score [268], have been developed. The fourth problem to mention is the approach that is utilized to determine the resiliency of cluster analysis findings. It is common for a strength measure to be based on a data aggregation score metric that is not utilized to determine the clustering algorithm present. Studying how the aforementioned options affect cluster analysis is an important part of understanding how findings may be generalized [269]. A more thorough understanding of biological processes is made possible by the data and information from diverse omics data types (also known as integrative genomics) and other heterogeneous information [270]. Methods of integrative omics data analysis have been classified into three categories [271]. In the first case, a lot of different types of data are combined and looked at at the same time. There are ways to look at the connections between metagenomics and metabolomics data with the help of CCA, according to the previous part of the text. People who study some types of data may be able to help people who study other types of data, which is called "data to knowledge." Using the example of colon cancer patients, a metagenomics study may lead to the discovery of possible genes that can be studied further with specialized proteome analysis tools. Finally, a sort of knowledge-to-knowledge analysis is one in which the data types are initially assessed individually, but the knowledge gathered is then merged to either identify hypotheses that are supported by several data types or provide a more thorough understanding of an individual event. To figure out how Crohn's disease starts and how it progresses, you can combine information from people with the disease who have a lot of extra genes and a lot of metabolic products in their digestive systems. This way, you can figure out how the disease starts and where it goes.

Overview of intelligent recommendation engine

One of the most important factors affecting our well-being is what is referred to as the "code of human biological processes" which is our microbiome. In 2022 M Prochazkova [272], supposed that The bacterial genome, in contrast to the human genome, is recognized for its protective factors, and as a result, it has a high degree of flexibility, giving numerous prospects for the discovery and development of new aspects of food, medicinal treatments, and nutrition modifications. Transferring from broad-sense dietary advice to guidance that takes into account your microbiome has not yet been successfully done, even though significant developments in microbiome data analysis [273]. It has not yet been possible to transition from population-wide dietary advice to Microbiome-aware nutritional recommendations, despite recent advancements in Microbiome research. To get a representative compilation of current Microbiome-aware nutrition recommendation exploration, look no further than Table 4. Following the identification of a tailored healthy target Microbiome employing data analytics approaches, a recommendation engine may use this knowledge to propose a way of creating it in the host and securing the health advantages. In the field of genomics, metagenomics is a hybrid term made up of the phrases meta and genomics. Consequently, genomics is the process of discovering the DNA sequence of an individual, whereas "meta" likes to refer to the fact that we are accomplishing it over a bunch of different unique species. Furthermore, metagenomics is often used for the investigation of microbial populations when it is hard to distinguish one microbe from another. For example, microscopic organisms may coexist in the same microenvironment, and their DNA fragments will be mixed together in the same consequence [274-277]. One technique for resolving the issue is the use of knowledge-based recommendation engines, in which suggestions are created using a limited number of approved drugs and nutritional therapeutic strategies. Even if this would be a wonderful starting point, the ability of such a system to give precise and tailored ideas, as is frequently the case with a platform that is capable of producing new commodities or processes on an individual basis, would be severely hampered. Current research indicates that [278, 279]. Even though such mechanistic modeling is very promising, it is currently hampered by several issues, including a lack of adequate characterization of an individual's gut and the metabolic pathways of their microbiome (microbiome). The application of artificial intelligence-based recommendation engines to food, drug design, and health is still in its early stages, despite a substantial amount of research on the subject [280-383]. This is an area that businesses are already getting into. Companies such as DAYTWO https://www.daytwo.com/ use 16S rRNA technology to give us information about our microbiota and give us nutritional advice. An example of a recommendation system is "any system that steers a user in a tailored form toward interesting and informative items in the huge space of stakeholders with different, or that gives intriguing or useful objects as an output" [284]. We'll talk about knowledge-based content-based and collaborative filtering strategies in the next part to get microbiome-aware nutritional advice.

Table 4 A few studies that talked about how to eat a nutritious diet for your microbiota composition were spotlighted

Interpretation of the data analysis	Factors associated with diet and eating	Metagenomic Data Processing Using High-Throughput Sequencing*	Reference
The algorithm takes into account personal, microbiome, and dietary information to choose the healthiest diet for individuals with diabetes Complications who want to lower their post-meal insulin sensitivity.	A person needs both phyto-nutrients and complex carbohydrates to stay healthy.	Microarray analysis of the 16S rRNA gene and full metagenomic analysis	[273]
To accurately predict how a person will react to various types of food products “many categories of flour”, the microbiome has a lot to do with how well it can do this.	a sort of sourdough	Microchip evaluation of the amplicons and comprehensive genomic investigation	[274]
The ability to efficiently predict how much weight the rodents will gain if they eat a normal diet or a high fat foods one is possible. "A eating plan where at least 35% of the total calories come from fats, both trans fatty acids and overloaded," says the report.	assessing gut microbiome empirical evidence on the relationship illness in the family	16S ribosomal RNA (rRNA) sequencing is used to look at how many different kinds of bacteria live in a place.	[275]
In computational prediction of recreated cellular metabolism through intestinal microbiota is used to provide individualized metabolic pathway supplemental prescriptions for Crohn 's syndrome.	dietary additives that support metabolic function	Genome-wide metagenomics	[276]
Using in silico modelling of metabolic pathways from the gut microbiome and host cells, the effects of high protein intake on stool samples and amino acid excretion levels are predicted given nutritionally important nutrients.	Carbohydrates, lipids, and protein are the three macronutrients that make up the majority of any diet.	16S ribosomal RNA (rRNA)	[277]

Information and Evidence Recommendation Engines create suggestions for improvement based on prior data

Ideal information and evidence the recommendation engine would be based on in silico models capable of properly modeling a certain person's gut. Comprehensive characterizations of the gut microbiota, human intestinal cells, intestinal and dietary metabolite concentrations, their interactions through metabolic pathways, and realistic target functions, among other things, are required for simulating such complex dynamics. An information-and evidence-based recommendation engine was designed and produced in a study including Crohn's patients and healthy control subjects [285]. It was discovered that after metagenomic data processing in the Rstudio 'BacArena' (https://cran.r-project.org/package = BacArena) [286, 287] &URL https://BacArena.github.io/, the researchers combined individual microbiome abundances with high - throughput sequencing metagenomic recreations of 818 microbiota cultivars from The Virtual Metabolic Human database “VMH database” [288], following metagenomic data processing in the R package BacArena. Using their molecular docking algorithms, which provide tailored metabolic supplements, it may be possible to enhance the short-chain fatty acid levels of thousands of patients. Preliminary studies on a smaller scale resulted in the development of a metabolic model of the gut microbiota, which is currently being applied [289]. For a more in-depth look at the subject, It's important to read D Hinojosa-Nogueira [290], and T Blasco [291], who write about it 2021, Knowledge-based decision aids have a lot of potential, but there are challenges to overcome before they can be used in the best way possible. The first obstacle to overcome is the poor availability and accuracy of high-throughput sequencing metagenomic recreations for gut microorganisms in the scientific community. Even though only 818 bacteria have high-throughput sequencing metagenomic recreations, according to recent research [292], 1,520 distinct microbes have been found in the human gut [293]. In-vitro simulations show some problems with the high-throughput sequencing metagenomic reconstruction of the gut microbiota. However, work is being done to close the gap between these two types of simulations. It's the second thing that needs to be fixed: the metabolic properties of the medium in your gastrointestinal system where your gut microorganisms thrive. If you want to do this, you need to do a very detailed analysis of the food metabolites that are available to microorganisms at different places in the gut. To do this, you need to be very careful with your dietary data. Also, the amount of time it takes to run in silico simulations gets longer as the number and complexity of high-throughput sequencing metagenomic reconstructions grows for both the host and the microbial species that live on it. Even though the list of problems might get longer, this paper isn't going to talk about them.

Evidenced Recommendation Engines are a kind of recommending software that involves information to provide predictions.

Alternatively, with Evidenced Recommendation Engines, frontiers in recommendations are dependent on the interests of other users for each item, although this is not the case in collaborative filtering Evidenced Recommendation Engines. They want to lower blood sugar levels after meals by using a tool called an "Evidenced Recommendation Engine" to make meal suggestions. This tool is called a "meal recommendation engine." First, each meal is put into groups based on how many calories it has. Then, the groups are broken down even more (macronutrients and micronutrients). Then, a regression model is created to predict post-meal glucose levels using the nutritional profile of the meal, the individual's microbiome features, and other personal information. The model, predicts post-meal glucose levels for each new person and each new meal. The meal that is most likely to have the lowest post-meal glucose level is recommended to the person based on these predictions. In a subsequent investigation, the same process is utilized, but this time using just [294]. It was discovered that a bread type recommendation system may be used to predict post-meal glucose levels in persons based on their Microbiome traits [295]. When developing Evidenced Recommendation Engines, a slew of challenges occur, which are described in more depth further down the page. The first barriers to overcome are unpredictability in data quality and incompatibility with other systems. When a certain set of users (or objects) is overrepresented in the data, the predictive model is more highly skewed in the advantage of the users' preferred products. As a consequence, the quality of the suggestions will improve as a result of these modifications, which will be highly variable. The use of stratified sampling may be able to alleviate some of the effects of this issue. The second issue is the difficulty in generalizing and personalizing suggestions, which is particularly problematic when feature vectors are not useful for prediction (this challenge is also pertinent to the "misplaced amounts" challenge noted in Table 3). Rather than being formed a priori, latent properties in collaborative filtering rule sets are learned rather than constructed a priori (Evidenced Recommendation Engines). To overcome the inherent constraints of context-based Evidenced Recommendation Engines (and vice versa), hybrid Evidenced Recommendation Engines are meant to take advantage of sentiment classification Recommendation Engines [296].

Systems for collaborative filtering and recommendation

What your consumers have done in the past is a good indicator of how they feel about a product currently. Recommendations are made based on the idea that consumers who give comparable ratings to current items would have a similar rating profile for everything else. Matrix completion and other collaboration filtering techniques are used to filter out unwanted information [297]. To manage user-assigned ratings, they are first structured into a sparse matrix in which columns correspond to different objects and rows relate to different users. In circumstances when the majority of users have only assessed a few items, the majority of the matrix is left empty. Matrix completion is the process of filling in the gaps in the matrix as a result of the similarities observed between people and products. A full review may be found in H Ko. 2022 [298]. When it comes to making dietary recommendations that are host interactions, collaborative filtering Recommendation Engines have not yet been applied successfully. To demonstrate how it may be utilized, we will go through an example in this section. A framework with each column representing a food plan and each row representing an individual is shown below. A specific value that may reflect gut microbiome alpha diversity during the period that the user adhered to a given dietary plan is shown below. Assuming that each individual has only tried a few different food regimens, the vast majority of the matrix will be blank. Using matrix completeness, we may fill the virtual world with expected alpha diversities to generate a full array, which is useful in this situation. This may be used to provide dietary guidelines to a person to increase the variety of their intestinal microbiota. Some difficulties exist while using collaborative filtering Recommendation Engines. Getting new users up and running might be difficult because of the lack of data they have to work with. It's important to note that the suggestions are based on commonalities among previous users, while new users have not tried any of the things in the database. The second obstacle to overcome is the curse of three-dimensionality. In addition, as the number of items rises, the likelihood of having user ratings for the same item configurations reduces; resulting in items and users being equally distinct (this is also important to the "high dimensionality" difficulties in Table 3). A hybrid Recommendation Engine may be employed in these situations. Following that, we'll go through a few hypothetical situations.

The following are a few examples of what could happen

We talked about a lot of different ways to use data analytics and recommendation systems to find out about people's Microbiomes and change their diets, as shown in Figures 1 and 3. The applicability of each approach is determined by the study goals and the availability of relevant data. In this section, we provide specific situations that serve as blueprints for merging key approaches into a single pipeline. The purpose of module 1 is to discover metabolic pathways that are enriched in the gut microbiome of healthy people by analyzing 16S rRNA data. The primary purpose of model 2 is to ensure that the necessary probiotic consumption is met to maintain a healthy gut microbiota. Initially, the participants would be characterized depending on the probiotic cultures they use (each carrying a distinct operational taxonomic unit) as well as their gut microbiota. Following that, each participant's microbiome scores will be generated based on the distance between the overloaded routes of their microbiome and the target healthy microbiome. A regression model must first be developed to predict microbiome scores based on intakes of operational taxonomic units. Finally, the suggested probiotic intake would be based on the operational taxonomic unit intake concentration expected to have an optimum microbiome score. In module 3, the solution is to determine the best foods for well-being, achievement, and illnesses. To build a compendium, you need to follow a consistent data collection and processing pipeline for the people who take part in the study. Using the compendium, researchers may create machine learning models to predict health indicators such as post-meal glucose levels or post-dieting weight regain [299]. The prediction models may then be utilized as the foundation of a recommendation system, determining the predicted influence of a certain diet on health for new people. A person's microbiota needs to be given metabolic supplements to make important compounds. This is case module 4. In the beginning, a metagenomic data processing pipeline is used to figure out the operational taxonomic units abundances of each person. Utilizing web technologies such as the Virtual Metabolic Human database, individual gut metabolic pathways are recreated. After everything is said and done, energy metabolism consumption restrictions for secreting important compounds of interest may be discovered by molecular docking models of high-throughput sequencing metagenomic recreations utilizing the Constraints Created Restoration Analysis (COBRA) method According to [300, 301], this empirically proven technique has only been used in a small range of empirical randomized trials.

Numerous innovations have occurred in the planetary system of intelligent property infringement

According to a significant increase in the number of issued patent lawsuits on the matter of the socialization between the colonic microbiota and dietary habits, as well as more extensively in the area of microbial community and public physical wellbeing, it is evident that investigations focused on the relationship between the gastrointestinal microbiota and specific diets have the potential to have a substantial influence on the application of the scientific method. More than 2,400 patents and trademarks related to the topics of "intestinal microbiota" and "nutrition" have been submitted by universities, hospitals, and private companies, including Microbiome Therapeutics (https://www.serestherapeutics.com/our-platform/), Gutguide (https://gutguide.com/), Whole Biome Inc. (https://www.easyleadz.com/company/whole-biome-inc/), and UBiome http://ubiome.com [302]. Despite the fact that some of these bestsellers were released within the last ten years, it is crucial to bear

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

in mind that this business is still in the very early stages of growth. This is something that must not be forgotten. Like the rapid growth in patent filings in this domain, the number of academic publications dedicated to the microbiome has increased substantially since 2007 [303]. A technique for modifying the prevalence of overweight and obesity (weight-to-size ratio) by significantly changing the nature of the microbial population is proposed in one of the oldest grant applications on the issue of microbiota and nutritious food that is publicly accessible (https://patents.google.com/patent/US20050239706A1/en?oq=US20050239706A1). Many more patents suggest utilizing the microbiota composition as a unique therapeutic, monitoring, and adjustment tool to alter the host phenotype, including but not limited to body weight reduction, bad lifestyle choices, and gastrointestinal disorders. Putting aside broad statements, preliminary invention applications in the disciplines of microbial population and weight management often address the topic of weight control by change of the intestinal microbiota (patents US20110123501A1 and US20100172874A1, respectively; https://patents.google.com/patent/US20110123501A1/en?oq=US20110123501A1 and https://patents.google.com/patent/US201001728 (patent number US20100172874A1, as well as patent portfolio number US20110123501A1). The development of novel probiotic microorganisms and the concomitant implementations are both described in intellectual property rights (Worldwide applications 2007136553A2), with some of the inventions trying to associate the bugs with certain target phenotypes such as weight control (European Patent Office 2178543B1, US9371510B2, US9113641B2, European Patent Office 2216036A1, European Patent Office 2296489A1, Worldwide applications 2010091991A1, and Worldwide applications 2010091991A1). Nestec SA, a business that undertakes research and gives consultancy services to Nestlé S.A., has filed a significant number of applications for probiotic weight reduction products via their website located at http://www.nestle.com/. As a consequence, patent applications linked to the gut microbiome of food have grown beyond obesity and weight management to encompass a number of other disorders as a consequence of further research on the relationship between host microbiome homeostasis and illness, as well as algorithms for large-scale data analysis. These developments have come about as a result of more studies on the link between host microbiome homeostasis and sickness. One of the many innovative uses of information about the gut microbiota and intake is in the diagnosis and treatment of problems in the proprioceptive framework; this is determined in part by analyzing information about the microbiota's composition (the United States 20170372027A1). Other implementations according to this standard include bioinformatics tools that are one-of-a-kind for comparing the locations of healthy young individuals and those who have gastrointestinal system pathological abnormalities (Worldwide applications). Use of microbial diversity and other techniques of data analysis to treat Crohn's disease, irritable bowel syndrome, ulcerative colitis, and gluten intolerance (US20170286620A1 United States https://patents.google.com/patent/US20170286620A1/en?oq=US20170286620A1)), and apparatus (equipment) US20170281091A1 the United States https://patents.google.com/There have also been a plethora of applications generated from research on the gastrointestinal system axis relationship, including analyzing and changing the gut flora to enhance memory or treat psychiatric diseases, among other things (Worldwide applications 2017171563A1 https://patents.google.com/patent/WO2017171563A1/en?oq=wo+2017171563A1 and Worldwide applications 2017160711A1 https://patents.google.com/patent/WO2017160711A1/en?oq=wo2017160711A1). According to a comprehensive review of patents associated with the microbiome, two recent advancements in the field include cancer diagnosis and therapy, as well as technology known as clumped regular basis preferring brief polynomial repetitions (CRISPR). In the past, several individuals have submitted patent applications for technologies that aim to exert control over the microbiota in the gut. However, regulatory guidelines have not kept pace with recent scientific discoveries. The European Food Safety Authority is responsible for ensuring the security of foods sold in Europe that are promoted as healthy. Another example of this would be probiotics. On the other hand, this is something that is overseen by the Food and Drug Administration in the United States of America. Laws and regulations are being updated in order to ensure that they are relevant to the dynamic nature of the industry. There is already a set of regulations, but the Food and Drug Administration has said that they will collaborate with the National Institutes of Health in the United States to ensure the public's safety (Food and Drug Administration, 2018). A product that is the result of biotechnology may be defined as "an active material other than a vaccine that includes germs and is meant to halt or relieve a communicable illness in living beings," according to the medical field. No probiotic currently on the market in the United States has been given marketing clearance by the Food and Drug Administration (Food and Drug Administration, 2016). According to the Food and Drug Administration, probiotics may now be sold in the United States as dietary supplements or as additions to food. However, manufacturers are not permitted to make claims that probiotics will cure, treat, or prevent any form of sickness since these kinds of claims are only permitted for pharmaceutical products. Because they do not include any live creatures, microbiome-targeted foods, which are also known as microbiota-directed nutrients, prebiotics, and dietary fiber, may be difficult to categorize. Other names for these foods include the aforementioned. Both pharmaceuticals and dietary supplements are under Food and Drug Administration jurisdiction, although the two categories are subject to separate regulations. Products that claim to be good for your health might fit into one of these two groups [304].

Conclusions

Imagine a new era of health and nutrition research in which food products can be made to work with the gut Microbiome in a predictable way to improve health. The integration of microbiology, genomics, analytical chemistry, machine learning, and information science has been made feasible by considerable developments in these fields. Many pioneering studies, including those listed below, have had encouraging findings, but the intricacy of the microbial community, the chemical characteristics of food, and their interplay in vitro and in vivo make substantial advancements in this specific subject challenging. There are new data sources that can be leveraged to solve this problem, such as electronic health records and developments in high-throughput sequencing for metabolomics assessment and dietary composition research. When it comes to this project, the use of computer-based science is now clear, because it will lay the groundwork for connecting all of these application packages and creating perspectives that can help us understand the complex interactions among both eating plans, gut Microbiomes, and general wellbeing better. This is a good thing. Incredibly changeable, the intestinal microbiota presents both challenges and opportunities for researchers. Because the Microbiome changes over time and between people, it's hard to find statistically significant signs that unambiguously show whether the microbiota is healthy or not. Because of its flexibility to adjust to changes in food and other environmental conditions, the gut Microbiome is a good target for diet-related therapies aimed at improving overall health. As illustrated in this quick review of current knowledge and possible research avenues, computational models science and data analytics software can play an important part in speeding research into the link between nutrition, microbiota, and public health. Computational models research and data analytics may play an essential role in accelerating research into the relationship between diet, gut Microbiome, and human health, as evidenced in this comprehensive introduction of scientific understanding and potential research tracks. It's possible to use these abilities to think about a future where certain communities or people might get a different type of food to help them get the best balance of good bacteria in their bodies. The system will first figure out which microbes are best for the target, the person, and the environment. You can get that targeting microbiota (discovery) through engineering. The second system will give you ideas on how to get there. Recognizing this difference and the need for smooth interaction between the two might strengthen collaborative research in this expanding sector. Since there has been such an increase in interest in studying the Microbiome, new software platforms and algorithms for processing and evaluating genomic sequencing data have appeared in recent years. Depending on the case, researchers may use a wide range of tools. They can use these tools to solve a wide range of problems. Many experts have shown that chemical-based datasets would be used to monitor the effectiveness of competing techniques based on the development's hypotheses. This gives relevant benchmarks. The growth of three-dimensional computer research may also result in the development of unique modeling and predictive analysis procedures that are individually suited towards each enterprise based on variables such as DNA sequencing, data requirements, computational complexity, and heterogeneity. There are ways to make standard procedures for things like the ones in big Tables 3 and 4. Food, the intestinal microbiota, and public health all collaborate, and we understand something and talk about it often, so we understand how much more we understand about it. People can use many different ways to figure out what makes an intestinal flora nutritious as well as what changes it. It is possible to have a dramatic impact on our everyday lives when recommendation algorithms are applied correctly. We believe that the next decade will be the era of computational nutrition, in which significant advancements in sequencing and computational tools will transform our relationship with food and diet.

References

1. Anwar H, Irfan S, Hussain G, et al. Gut Microbiome: A New Organ System in Body. *Parasitol Microbiol Res* 2020. http://doi.org/10.5772/intechopen.89634
2. Schloss PD, Westcott SL, Ryabin T, et al. Introducing mothur: Open-Source, Platform-Independent, Community-Supported Software for Describing and Comparing Microbial Communities. *Appl Environ Microbiol* 2009;75(23):7537–7541. http://doi.org/10.1128/AEM.01541-09
3. Fenech M, El-Sohemy A, Cahill L, et al. Nutrigenetics and Nutrigenomics: Viewpoints on the Current Status and Applications in Nutrition Research and Practice. *J Nutrigenet Nutrigenomics* 2011;4(2):69–89. http://doi.org/10.1159/000327772
4. Mierziak J, Kostyn K, Boba A, Czemplik M, Kulma A, Wojtasik W. Influence of the Bioactive Diet Components on the Gene Expression Regulation. *Nutrients* 2021;13(11):3673. http://doi.org/10.3390/nu13113673
5. Barratt MJ, Lebrilla C, Shapiro H-Y, Gordon JI. The Gut Microbiota, Food Science, and Human Nutrition: A Timely Marriage. *Cell Host Microbe* 2017;22(2):134–141. http://doi.org/10.1016/j.chom.2017.07.006
6. Wishart DS. Emerging applications of metabolomics in drug discovery and precision medicine. *Nat Rev Drug Discov* 2016;15(7):473–484. http://doi.org/10.1038/nrd.2016.32
7. Hulsen T, Jamuar SS, Moody AR, et al. From Big Data to Precision Medicine. *Front Med* 2019;6:34. http://doi.org/10.3389/fmed.2019.00034
8. Cullen CM, Aneja KK, Beyhan S, et al. Emerging Priorities for Microbiome Research. *Front Microbiol* 2020;11:136. http://doi.org/10.3389/fmicb.2020.00136
9. Koponen KK, Salosensaari A, Ruuskanen MO, et al. Associations of healthy food choices with gut microbiota profiles. *Am J Clin Nutr* 2021;114(2):605–616. http://doi.org/10.1093/ajcn/nqab077
10. Camacho DM, Collins KM, Powers RK, Costello JC, Collins JJ. Next-Generation Machine Learning for Biological Networks. *Cell* 2018;173(7):1581–1592. http://doi.org/10.1016/j.cell.2018.05.015
11. van Erp M, Reynolds C, Maynard D, et al. Using Natural Language Processing and Artificial Intelligence to Explore the Nutrition and Sustainability of Recipes and Food. *Front Artif Intell* 2021;3:621577. http://doi.org/10.3389/frai.2020.621577
12. Liu R, Li X, Lam KS. Combinatorial chemistry in drug discovery. *Curr Opin Chem Biol* 2017;38:117–126. http://doi.org/10.1016/j.cbpa.2017.03.017
13. Miller M, Zhu C, Bromberg Y. clubber: removing the bioinformatics bottleneck in big data analyses. *J Integr Bioinform* 2017;14(2):20170020. http://doi.org/10.1515/jib-2017-0020
14. King CH, Desai H, Sylvetsky AC, et al. Baseline human gut microbiota profile in healthy people and standard reporting template. *PLoS ONE* 2019;14(9):e0206484. http://doi.org/10.1371/journal.pone.0206484
15. Baothman OA, Zamzami MA, Taher I, Abubaker J, Abu-Farha M. The role of Gut Microbiota in the development of obesity and Diabetes. *Lipids Health Dis* 2016;15:108. http://doi.org/10.1186/s12944-016-0278-4
16. Evert AB, Dennison M, Gardner CD, et al. Nutrition Therapy for Adults With Diabetes or Prediabetes: A Consensus Report. *Diabetes Care* 2019;42(5):731–754. http://doi.org/10.2337/dci19-0014
17. Szczerba M, Wiewiórka MS, Okoniewski MJ, Rybiński H. Scalable Cloud-Based Data Analysis Software Systems for Big Data from Next Generation Sequencing. In: Japkowicz, N., Stefanowski, J. (eds) Big Data Analysis: New Algorithms for a New Society. *Stud Big Data* 2016;16:263–283. http://doi.org/10.1007/978-3-319-26989-4_11
18. Johnson AJ, Zheng JJ, Kang JW, Saboe A, Knights D, Zivkovic AM. A Guide to Diet-Microbiome Study Design. *Front Nutr* 2020;7:79. http://doi.org/10.3389/fnut.2020.00079
19. Zmora N, Suez J, Elinav E. You are what you eat: diet, health and the gut microbiota. *Nat Rev Gastroenterol Hepatol* 2018;16(1):35–56. http://doi.org/10.1038/s41575-018-0061-2
20. Roager HM, Dragsted LO. Diet-derived microbial metabolites in health and disease. *Nutr Bull* 2019;44(3):216–227. http://doi.org/10.1111/nbu.12396
21. Sherwin E, Dinan TG, Cryan JF. Recent developments in understanding the role of the gut microbiota in brain health and disease. *Ann NY Acad Sci* 2017;1420(1):5–25. http://doi.org/10.1111/nyas.13416
22. Cruz-Pereira JS, Rea K, Nolan YM, O'Leary OF, Dinan TG, Cryan JF. Depression's Unholy Trinity: Dysregulated Stress, Immunity, and the Microbiome. *Annu Rev Psychol* 2020;71(1):49–78. http://doi.org/10.1146/annurev-psych-122216-011613
23. Stinson LF, Boyce MC, Payne MS, Keelan JA. The Not-so-Sterile Womb: Evidence That the Human Fetus Is Exposed to Bacteria Prior to Birth. *Front Microbiol* 2019;10:1124. http://doi.org/10.3389/fmicb.2019.01124
24. Granger CL, Embleton ND, Palmer JM, Lamb CA, Berrington JE, Stewart CJ. Maternal breastmilk, infant gut microbiome and the impact on preterm infant health. *Acta Paediatr* 2020;110(2):450–457. http://doi.org/10.1111/apa.15534
25. Beck LC, Granger CL, Masi AC, Stewart CJ. Use of omic technologies in early life gastrointestinal health and disease: from bench to bedside. *Expert Rev Proteomics* 2021;18(4):247–259. http://doi.org/10.1080/14789450.2021.1922278
26. Sarkar A, Yoo JY, Valeria Ozorio Dutra S, Morgan KH, Groer M. The Association between Early-Life Gut Microbiota and Long-Term Health and Diseases. *J Clin Med* 2021;10(3):459. http://doi.org/10.3390/jcm10030459
27. Gregora M. *Lactobacillus* Species in Breast Milk. *Probiotics - Curr*

Knowl Future Prospects 2018. http://doi.org/10.5772/intechopen.72639
28. Reyed RM. The short-term impact of probiotic intervention on the human gut microbiome facilitates fermentation potential of indigestible carbohydrates to short- chain fatty acids for preventing and treatment of obesity. *J Food Clin Nut Vol* 2021;4(1):1. https://www.pulsus.com/scholarly-articles/the-shortterm-impact-of-probiotic-intervention-on-the-human-gut-microbiome-facilitate-fermentation-potential-of-indigest.pdf
29. Reyed R, El-Diwany A. Molasses as bifidus promoter on bifidobacteria and lactic acid bacteria growing in skim milk. *Internet J Microbiol* 2007;5(1):1–8. https://ispub.com/IJMB/5/1/10975
30. R Reyed. Isolation, identification of human autochthonous bifidobacteria and comparison of its growth on different natural food products. *Internet J Microbiol* 2007;3(2). https://www.semanticscholar.org/paper/Isolation%2C-Identification-Of-Human-Autochthonous-Of-Reyed/03eac1c1073d1e80fdc382dd0d581b56020e1262
31. Reyed RM. Probiotics: a new strategies for prevention and therapy of diarrhea disease. *J Appl Sci Res* 2007;3(4):291–299. http://www.aensiweb.com/old/jasr/jasr/2007/291-299.pdf
32. Reyed RM. The role of bifidobacteria in health. *Res J Medicine & Med Sci* 2007;2(1):14–24. https://www.researchgate.net/profile/Reyed-M/publication/228635134_The_Role_of_Bifidobacteria_in_Health/links/5b3b2850aca2720785055499/The-Role-of-Bifidobacteria-in-Health.pdf
33. Taniguchi M, Nambu M, Katakura Y, Yamasaki-Yashiki S. Adhesion mechanisms of *Bifidobacterium animalis* subsp. *lactis* JCM 10602 to dietary fiber. *Biosci Microbiota Food Health* 2021;40(1):59–64. http://doi.org/10.12938/bmfh.2020-003
34. Li SC, Hsu WF, Chang JS, Shih CK. Combination of *Lactobacillus acidophilus* and *Bifidobacterium animalis* subsp. lactis Shows a Stronger Anti-Inflammatory Effect than Individual Strains in HT-29 Cells. *Nutrients* 2019;11(5):969. http://doi.org/10.3390/nu11050969
35. Saturio S, Nogacka AM, Alvarado-Jasso GM, et al. Role of Bifidobacteria on Infant Health. *Microorganisms* 2021;9(12):2415. http://doi.org/10.3390/microorganisms9122415
36. Iddrisu I, Monteagudo-Mera A, Poveda C, et al. Malnutrition and Gut Microbiota in Children. *Nutrients* 2021;13(8):2727. http://doi.org/10.3390/nu13082727
37. Bäckhed F, Roswall J, Peng Y, et al. Dynamics and Stabilization of the Human Gut Microbiome during the First Year of Life. *Cell Host Microbe* 2015;17(5):690–703. http://doi.org/10.1016/j.chom.2015.04.004
38. Bottacini F, Ventura M, van Sinderen D, O'Connell Motherway M. Diversity, ecology and intestinal function of bifidobacteria. *Microb Cell Fact* 2014;13(S1):S4. http://doi.org/10.1186/1475-2859-13-S1-S4
39. Yeoh YK, Zuo T, Lui GC-Y, et al. Gut microbiota composition reflects disease severity and dysfunctional immune responses in patients with COVID-19. *Gut* 2021;70(4):698–706. http://doi.org/10.1136/gutjnl-2020-323020
40. Zuo T, Liu Q, Zhang F, et al. Depicting SARS-CoV-2 faecal viral activity in association with gut microbiota composition in patients with COVID-19. *Gut* 2021;70(2):276–284. http://doi.org/10.1136/gutjnl-2020-322294
41. Bozkurt HS, Quigley EM. The probiotic Bifidobacterium in the management of Coronavirus: A theoretical basis. *Int J Immunopathol Pharmacol* 2020;34:205873842096130. http://doi.org/10.1177/2058738420961304
42. Marras L, Caputo M, Bisicchia S, et al. The Role of Bifidobacteria in Predictive and Preventive Medicine: A Focus on Eczema and Hypercholesterolemia. *Microorganisms* 2021;9(4):836. http://doi.org/10.3390/microorganisms9040836
43. Food Standards Agency. Allergen guidance for food businesses: Guidance for food businesses on providing allergen information and best practice for handling allergens. https://www.food.gov.uk/business-guidance/allergen-guidance-for-food-businesses
44. Fluhr L, Mor U, Kolodziejczyk AA, et al. Gut microbiota modulates weight gain in mice after discontinued smoke exposure. *Nature* 2021;600(7890):713–719. http://doi.org/10.1038/s41586-021-04194-8
45. Tiew PY, Jaggi TK, Chan LLY, Chotirmall SH. The airway microbiome in COPD, bronchiectasis and bronchiectasis-COPD overlap. *Clin Respir J* 2020;15(2):123–133. http://doi.org/10.1111/crj.13294
46. Thompson RS, Vargas F, Dorrestein PC, Chichlowski M, Berg BM, Fleshner M. Dietary prebiotics alter novel microbial dependent fecal metabolites that improve sleep. *Sci Rep* 2020;10(1):3848. http://doi.org/10.1038/s41598-020-60679-y
47. McDonald D, Hyde E, Debelius JW, et al. American Gut: an Open Platform for Citizen Science Microbiome Research. *mSystems* 2018;3(3):e00031-18. https://doi.org/10.1128/msystems.00031-18
48. Nikolova VL, Smith MRB, Hall LJ, Cleare AJ, Stone JM, Young AH. Perturbations in Gut Microbiota Composition in Psychiatric Disorders. *JAMA Psychiatry* 2021;78(12):1343. http://doi.org/10.1001/jamapsychiatry.2021.2573
49. Bull MJ, Plummer NT. Part 1: The Human Gut Microbiome in Health and Disease. *Integr Med (Encinitas)*. 2014;13(6):17–22. http://www.ncbi.nlm.nih.gov/pmc/articles/pmc4566439/
50. Pascal V, Pozuelo M, Borruel N, et al. A microbial signature for Crohn's disease. *Gut* 2017;66(5):813–822. http://doi.org/10.1136/gutjnl-2016-313235
51. Ramirez J, Guarner F, Bustos Fernandez L, Maruy A, Sdepanian VL, Cohen H. Antibiotics as Major Disruptors of Gut Microbiota. *Front Cell Infect Microbiol* 2020;10:572912. http://doi.org/10.3389/fcimb.2020.572912
52. Johny TK, Puthusseri RM, Bhat SG. A primer on metagenomics and next-generation sequencing in fish gut microbiome research. *Aquac Res* 2021;52(10):4574–4600. http://doi.org/10.1111/are.15373
53. Wooley JC, Godzik A, Friedberg I. A Primer on Metagenomics. *PLoS Comput Biol* 2010;6(2):e1000667. http://doi.org/10.1371/journal.pcbi.1000667
54. Bauermeister A, Mannochio-Russo H, Costa-Lotufo LV, Jarmusch AK, Dorrestein PC. Mass spectrometry-based metabolomics in microbiome investigations. *Nat Rev Microbiol* 2021;20(3):143–160. http://doi.org/10.1038/s41579-021-00621-9
55. Wu G, Zhao N, Zhang C, Lam YY, Zhao L. Guild-based analysis for understanding gut microbiome in human health and diseases. *Genome Med* 2021;13:22. http://doi.org/10.1186/s13073-021-00840-y
56. Graf AC, Striesow J, Pané-Farré J, et al. An Innovative Protocol for Metaproteomic Analyses of Microbial Pathogens in Cystic Fibrosis Sputum. *Front Cell Infect Microbiol* 2021;11:724569. http://doi.org/10.3389/fcimb.2021.724569
57. Herold M, Martínez Arbas S, Narayanasamy S, et al. Integration of time-series meta-omics data reveals how microbial ecosystems respond to disturbance. *Nat Commun* 2020;11:5281. http://doi.org/10.1038/s41467-020-19006-2
58. Xu J, Yang Y. Gut microbiome and its meta-omics perspectives: profound implications for cardiovascular diseases. *Gut Microbes* 2021;13(1):1936379. http://doi.org/10.1080/19490976.2021.1936379
59. Wang J, Zhang J, Liu W, Zhang H, Sun Z. Metagenomic and metatranscriptomic profiling of Lactobacillus casei Zhang in the human gut. *npj Biofilms Microbiomes* 2021;7:55. http://doi.org/10.1038/s41522-021-00227-2

60. Li D, Gao C, Zhang F, et al. Seven facts and five initiatives for gut microbiome research. *Protein Cell* 2020;11(6):391–400. http://doi.org/10.1007/s13238-020-00697-8
61. Durazzi F, Sala C, Castellani G, Manfreda G, Remondini D, De Cesare A. Comparison between 16S rRNA and shotgun sequencing data for the taxonomic characterization of the gut microbiota. *Sci Rep* 2021;11(1). http://doi.org/10.1038/s41598-021-82726-y
62. Balvočiūtė M, Huson DH. SILVA, RDP, Greengenes, NCBI and OTT — how do these taxonomies compare? *BMC Genomics* 2017;18(S2):114. http://doi.org/10.1186/s12864-017-3501-4
63. Durazzi F, Sala C, Castellani G, Manfreda G, Remondini D, De Cesare A. Comparison between 16S rRNA and shotgun sequencing data for the taxonomic characterization of the gut microbiota. *Sci Rep* 2021;11(1):3030. http://doi.org/10.1038/s41598-021-82726-y
64. Konopiński MK. Shannon diversity index: a call to replace the original Shannon's formula with unbiased estimator in the population genetics studies. *PeerJ* 2020;8:e9391. http://doi.org/10.7717/peerj.9391
65. Bliek Tijs, Frans van der Kloet and Marc Galland (eds): (2020) "RNA-seq lesson." Version. 04. https://github.com/ScienceParkStudyGroup/rnaseq-lesson
66. Magurran AE. Measuring biological diversity. 2004;29(2):285–286. https://doi.org/10.2989/16085910409503825
67. Colwell RK, Elsensohn JE. EstimateS turns 20: statistical estimation of species richness and shared species from samples, with non-parametric extrapolation. *Ecography* 2014;37(6): 609-613. https://doi.org/10.1111/ecog.00814
68. Lozupone CA, Hamady M, Kelley ST, Knight R. Quantitative and Qualitative β Diversity Measures Lead to Different Insights into Factors That Structure Microbial Communities. *Appl Environ Microbiol* 2007;73(5):1576–1585. http://doi.org/10.1128/AEM.01996-06
69. Ivanov I. Gut-host Crosstalk: Methodological and Computational Challenges. *Dig Dis Sci* 2020;65(3):686-694. http://doi.org/10.1007/s10620-020-06105-9
70. Lozupone C, Knight R. UniFrac: a New Phylogenetic Method for Comparing Microbial Communities. *Appl Environ Microbiol* 2005;71(12):8228–8235. http://doi.org/10.1128/AEM.71.12.8228-8235.2005
71. Hamady M, Lozupone C, Knight R. Fast UniFrac: facilitating high-throughput phylogenetic analyses of microbial communities including analysis of pyrosequencing and PhyloChip data. *ISME J* 2009;4(1):17–27. http://doi.org/10.1038/ismej.2009.97
72. Chen J, Bittinger K, Charlson ES, et al. Associating microbiome composition with environmental covariates using generalized UniFrac distances. *Bioinformatics* 2012;28(16):2106–2113. http://doi.org/10.1093/bioinformatics/bts342
73. Lozupone C, Hamady M, Knight R. UniFrac – An online tool for comparing microbial community diversity in a phylogenetic context. *BMC Bioinformatics* 2006;7(1):371. http://doi.org/10.1186/1471-2105-7-371
74. Whitfield-Cargile CM, Cohen ND, Suchodolski J, et al. Composition and Diversity of the Fecal Microbiome and Inferred Fecal Metagenome Does Not Predict Subsequent Pneumonia Caused by Rhodococcus equi in Foals. *PLoS ONE* 2015;10(8):e0136586. http://doi.org/10.1371/journal.pone.0136586
75. Horvath A, Leber B, Feldbacher N, et al. Effects of a multispecies synbiotic on glucose metabolism, lipid marker, gut microbiome composition, gut permeability, and quality of life in diabesity: a randomized, double-blind, placebo-controlled pilot study. *Eur J Nutr* 2019;59(7):2969–2983. http://doi.org/10.1007/s00394-019-02135-w
76. van Bommel EJM, Herrema H, Davids M, Kramer MHH, Nieuwdorp M, van Raalte DH. Effects of 12-week treatment with dapagliflozin and gliclazide on faecal microbiome: Results of a double-blind randomized trial in patients with type 2 diabetes. *Diabetes Metabolism* 2020;46(2):164–168. http://doi.org/10.1016/j.diabet.2019.11.005
77. Frost F, Storck LJ, Kacprowski T, et al. A structured weight loss program increases gut microbiota phylogenetic diversity and reduces levels of Collinsella in obese type 2 diabetics: A pilot study. *PLoS ONE* 2019;14(7):e0219489. http://doi.org/10.1371/journal.pone.0219489
78. Qian XB, Chen T, Xu YP, et al. A guide to human microbiome research: study design, sample collection, and bioinformatics analysis. *Chin Med J (Engl)* 2020;133(15):1844–1855. http://doi.org/10.1097/CM9.0000000000000871
79. Shao J, Tu D. *The Jackknife and Bootstrap*. New York: Springer-Verlag, 1995.
80. Chernick MR. *Bootstrap Methods: A Guide for Practitioners and Researchers*. 2nd Ed. Hoboken: John Wiley & Sons, Wiley Online Library Google Scholar, 2007.
81. Sharma P, Chetty P. How to perform bootstrap and jackknife analysis? Knowledge Tank, Project Guru. https://www.projectguru.in/bootstrap-jackknife-analysis/
82. King CH, Desai H, Sylvetsky AC, et al. Baseline human gut microbiota profile in healthy people and standard reporting template. *PLoS ONE* 2019;14(9):e0206484. http://doi.org/10.1371/journal.pone.0206484
83. Claesson MJ, Jeffery IB, Conde S, et al. Gut microbiota composition correlates with diet and health in the elderly. *Nature* 2012;488(7410):178–184. http://doi.org/10.1038/nature11319
84. Cotillard A, Kennedy SP, Kong LC, et al. Dietary intervention impact on gut microbial gene richness. *Nature* 2013;500(7464):585–588. https://doi.org/10.1038/nature12480
85. McDonald D, Hyde E, Debelius JW, et al. American Gut: An Open Platform for Citizen Science Microbiome Research. *MSystems* 2018;3(3):e00031-18. https://doi.org/10.1101/277970
86. Rinninella, Cintoni, Raoul, et al. Food Components and Dietary Habits: Keys for a Healthy Gut Microbiota Composition. *Nutrients* 2019;11(10):2393. http://doi.org/10.3390/nu11102393
87. Turnbaugh PJ, Ridaura VK, Faith JJ, Rey FE, Knight R, Gordon JI. The Effect of Diet on the Human Gut Microbiome: A Metagenomic Analysis in Humanized Gnotobiotic Mice. *Sci Transl Med* 2009;1(6):614. http://doi.org/10.1126/scitranslmed.3000322
88. Wu GD, Chen J, Hoffmann C, et al. Linking Long-Term Dietary Patterns with Gut Microbial Enterotypes. *Science* 2011;334(6052):105–108. http://doi.org/10.1126/science.1208344
89. Battaglioli EJ, Kashyap PC. Diet Effects on Gut Microbiome Composition, Function, and Host Physiology. *Physiology of the Gastrointestinal Tract* 2018:755–766. http://doi.org/10.1016/B978-0-12-809954-4.00033-5
90. FAO (2018) Dietary Assessment: A resource guide to method selection and application in low resource settings. Rome.
91. Schembre SM, Liao Y, O'Connor SG, et al. Mobile Ecological Momentary Diet Assessment Methods for Behavioral Research: Systematic Review. *JMIR Mhealth Uhealth* 2018;6(11):e11170. http://doi.org/10.2196/11170
92. Dao KP, De Cocker K, Tong HL, Kocaballi AB, Chow C, Laranjo L. Smartphone-Delivered Ecological Momentary Interventions Based on Ecological Momentary Assessments to Promote Health Behaviors: Systematic Review and Adapted Checklist for Reporting Ecological Momentary Assessment and Intervention Studies. *JMIR Mhealth Uhealth* 2021;9(11):e22890. http://doi.org/10.2196/22890

93. Hinojosa-Nogueira D, Pérez-Burillo S, Navajas-Porras B, et al. Development of an Unified Food Composition Database for the European Project "Stance4Health". *Nutrients* 2021;13(12):4206. http://doi.org/10.3390/nu13124206
94. Snetselaar LG, de Jesus JM, DeSilva DM, Stoody EE. Dietary Guidelines for Americans, 2020–2025. *Nutr Today* 2021;56(6):287–295. http://doi.org/10.1097/NT.0000000000000512
95. Bhagwat, Seema; Haytowitz, David B. USDA Database for the Flavonoid Content of Selected Foods. Release 3.2 (November 2015). https://doi.org/10.15482/USDA.ADC/1324465
96. Bhagwat S, Haytowitz DB, Holden JM. *USDA Database for the Isoflavone Content of Selected Foods, Release 2.0*. Maryland: US Department of Agriculture; 2008.
97. Rothwell JA, Urpi-Sarda M, Boto-Ordoñez M, et al. Systematic analysis of the polyphenol metabolome using the Phenol-Explorer database. *Mol Nutr Food Res* 2015;60(1):203–211. http://doi.org/10.1002/mnfr.201500435
98. Plumb J, Pigat S, Bompola F, et al. eBASIS (Bioactive Substances in Food Information Systems) and Bioactive Intakes: Major Updates of the Bioactive Compound Composition and Beneficial Bioeffects Database and the Development of a Probabilistic Model to Assess Intakes in Europe. *Nutrients* 2017;9(4):320. http://doi.org/10.3390/nu9040320
99. U S Food and Drug Administration, Department of Health and Human Services (2008) Food Labeling. Code of Federal Regulations, Title 21, Pt. 101. U.S. Government Printing Office website: https://www.fda.gov/files/food/published/Food-Labeling-Guide-%28PDF%29.pdf
100. Institute of Medicine (US) Committee on Examination of Front-of-Package Nutrition Rating Systems and Symbols. Front-of-Package Nutrition Rating Systems and Symbols: Phase I Report. Wartella EA, Lichtenstein AH, Boon CS, editors. Washington (DC): National Academies Press (US); 2010. https://pubmed.ncbi.nlm.nih.gov/24983042/
101. GRIN-Global. U.S. National Plant Germplasm System. https://npgsweb.ars-grin.gov/gringlobal/search.aspx Accessed on 25 March 2022
102. U.S. Department of Agriculture (2007) USDA Table of Nutrient Retention Factors, Release 6. Nutrient Data Laboratory Web Site: http://www.ars.usda.gov/Services/docs.htm?docid=9448 Accessed June 25, 2011.
103. U.S. Department of Agriculture (1984) Oxalic Acid Content of Selected Vegetables. Nutrient Data Laboratory Web Site: http://www.ars.usda.gov/Services/docs.htm?docid=9444 Accessed June 25, 2011.
104. Chazelas E, Deschasaux M, Srour B, et al. Food additives: distribution and co-occurrence in 126,000 food products of the French market. *Sci Rep* 2020;10(1):3980. http://doi.org/10.1038/s41598-020-60948-w
105. FoodOn. About the FoodOn Consortium (2021). https://foodon.org/about/ Accessed March 24, 2021.
106. Church SM. The history of food composition databases. *Nutr Bulletin* 2006;31(1):15–20. http://doi.org/10.1111/j.1467-3010.2006.00538.x
107. From Science to Practice: Research and Knowledge to Achieve the SDGs. Policy Brief, UNRIDS, Geneve, Switzerland. www.unrisd.org/rpb38 Published July 3, 2021; Accessed March 24, 2021.
108. Bawajeeh A, Kalendar S, Scarpa G, et al. Development of an Arabic food composition database for use in an Arabic online dietary assessment tool (myfood24). *J Food Compos Anal* 2021;102:104047. http://doi.org/10.1016/j.jfca.2021.104047
109. Tayel DI, Aboudeif NM. Evaluation of Dietary Intake Analysis Using Egyptian Modified Food Processor Software and Traditional Method: A Comparative Study. *Can J Clin Nutr* 2020;8(1):36–53. http://doi.org/10.14206/canad.j.clin.nutr.2020.01.04
110. Lupiañez-Barbero A, González Blanco C, de Leiva Hidalgo A. Tablas y bases de datos de composición de alimentos españolas: necesidad de un referente para los profesionales de la salud. *Endocrinol Diabetes Nutr* 2018;65(6):361–73. http://doi.org/10.1016/j.endinu.2018.05.001
111. Mitsopoulou AV, Magriplis E, Dimakopoulos I, et al. Micronutrient intakes and their food sources among Greek children and adolescents. *Public Health Nutr* 2020;23(13):2314–2326. http://doi.org/10.1017/S136898001900449X
112. Grande F. Reformulation of the Brazilian Food Composition Table (TBCA) database. Inter-Unit Graduate Studies in Applied Human Nutrition - PRONUT, FCF/FSP/FEA - USP, São Paulo, 2018. http://www.teses.usp.br/teses/disponiveis/89/89131/tde-12112018-152540/es.php
113. FoodDB Version 1.0. 2021. www.foodb.ca Accessed March 10, 2021.
114. Delgado A, Issaoui M, Vieira MC, Saraiva de Carvalho I, Fardet A. Food Composition Databases: Does It Matter to Human Health? *Nutrients* 2021;13(8):2816. http://doi.org/10.3390/nu13082816
115. Durazzo A, Lucarini M. Food Composition and Dedicated Databases: Key Tools for Human Health and Public Nutrition. *Nutrients* 2021;13(11):4003. http://doi.org/10.3390/nu13114003
116. Kingston C, Kehoe L, McNulty BA, et al. Updating of the Irish Food Composition Database for vitamin K1 and vitamin K2. *Proc Nutr Soc* 2020;79(OCE1):E24. http://doi.org/10.1017/S0029665119001459
117. Delgado A, Issaoui M, Vieira MC, Saraiva de Carvalho I, Fardet A. Food Composition Databases: Does It Matter to Human Health? *Nutrients* 2021;13(8):2816. http://doi.org/10.3390/nu13082816
118. Paul A, Southgate D, Russell J. *First supplement to McCance and Widdowson's The Composition of Foods Amino Acid composition (per 100g foods), Fatty Acid composition (per 100g foods)*. London: HMSO; 1980.
119. Institute of Nutrition, Mahidol University (2014). ASEAN Food Composition Database, Electronic version 1, February 2014, Thailand. http://www.inmu.mahidol.ac.th/aseanfoods/composition_data.html
120. Mendoza R, Tolentino-Mayo L, Hernández-Barrera L, Nieto C, Monterrubio-Flores E, Barquera S. Modifications in the Consumption of Energy, Sugar, and Saturated Fat among the Mexican Adult Population: Simulation of the Effect When Replacing Processed Foods that Comply with a Front of Package Labeling System. *Nutrients* 2018;10(1):101. http://doi.org/10.3390/nu10010101
121. Janowska-Miasik E, Waśkiewicz A, Witkowska AM, et al. Diet quality in the population of Norway and Poland: differences in the availability and consumption of food considering national nutrition guidelines and food market. *BMC Public Health* 2021;21(1):319. http://doi.org/10.1186/s12889-021-10361-3
122. Tabela de Composição de Alimentos (TCA). Instituto Nacional de Saúde Doutor Ricardo Jorge, Lisboa, Version 3.2–2018. http://portfir.insa.pt/foodcomp/search Accessed October 4, 2019
123. Federal Food Safety and Veterinary Office, Swiss Society for Nutrition. https://www.blv.admin.ch/blv/en/home.html
124. Dietary Guidelines for the Brazilian population, translated by Carlos Augusto Monteiro. Brasília: Ministry of Health of Brazil, 2015. https://bvsms.saude.gov.br/bvs/publicacoes/dietary_guidelines

_brazilian_population.pdf
125. Montville JB, Ahuja JKC, Martin CL, et al. USDA Food and Nutrient Database for Dietary Studies (FNDDS), 5.0. *Procedia Food Sci* 2013;2:99–112. http://doi.org/10.1016/j.profoo.2013.04.016
126. Turkish Food Composition Database. http://www.turkomp.gov.tr/main Accessed March 11, 2021.
127. Khalis M, Garcia-Larsen V, Charaka H, et al. Update of the Moroccan food composition tables: Towards a more reliable tool for nutrition research. *J Food Compos Anal* 2020;87:103397. http://doi.org/10.1016/j.jfca.2019.103397
128. Wishart DS, Guo A, Oler E, et al. HMDB 5.0: the Human Metabolome Database for 2022. *Nucleic Acids Res* 2022;7;50(D1):D622–D631. https://doi.org/10.1093/nar/gkab1062
129. Wishart DS, Tzur D, Knox C, et al. HMDB: the Human Metabolome Database. *Nucleic Acids Res* 2007;35(Database):D521–26. http://doi.org/10.1093/nar/gkl923
130. Wishart DS, Feunang YD, Marcu A, et al. HMDB 4.0: the human metabolome database for 2018. *Nucleic Acids Res* 2017;46(D1):D608–17. http://doi.org/10.1093/nar/gkx1089
131. Wishart DS, Knox C, Guo AC, et al. HMDB: a knowledgebase for the human metabolome. *Nucleic Acids Res* 2009;37(Database):D603–10. http://doi.org/10.1093/nar/gkn810
132. Wishart DS, Jewison T, Guo AC, et al. HMDB 3.0—The Human Metabolome Database in 2013. *Nucleic Acids Res* 2012;41(D1):D801–7. http://doi.org/10.1093/nar/gks1065
133. Mitchell AL, Almeida A, Beracochea M, et al. MGnify: the microbiome analysis resource in 2020. *Nucleic Acids Res* 2020 Jan 8;48(D1):D570–D578. http://doi.org/10.1093/nar/gkz1035
134. Keller-Costa T, Lago-Lestón A, Saraiva JP, et al. Metagenomic insights into the taxonomy, function, and dysbiosis of prokaryotic communities in octocorals. *Microbiome* 2021;9(1):72. http://doi.org/10.1186/s40168-021-01031-y
135. Lopes I, Altab G, Raina P, de Magalhães JP. Gene Size Matters: An Analysis of Gene Length in the Human Genome. *Front Genet* 2021;12:559998. http://doi.org/10.3389/fgene.2021.559998
136. EFSA Panel on Nutrition, Novel foods and Food allergens (NDA); Turck D, Castenmiller J, De Henauw S, et al. Beta-glucans from oats and/or barley in a ready-to-eat cereal manufactured via pressure cooking and reduction of blood-glucose rise after consumption: evaluation of a health claim pursuant to Article 13(5) of Regulation (EC) No 1924/2006. *EFSA J* 2021 Apr 8;19(4):e06493. https://doi.org/10.2903/j.efsa.2021.6493
137. Myhrstad MCW, Slydahl M, Hellmann M, et al. Nutritional quality and costs of gluten-free products: a case-control study of food products on the Norwegian marked. *Food Nutr Res* 2021;65. http://doi.org/10.29219/fnr.v65.6121
138. Giannakourou MC, Saltaouras K, Stoforos NG. On optimum dynamic temperature profiles for thermal inactivation kinetics determination. *J Food Sci* 2021;86(6):2172–2193. http://doi.org/10.1111/1750-3841.15770
139. Racchi I, Scaramuzza N, Hidalgo A, Berni E. Combined effect of water activity and pH on the growth of food-related ascospore-forming molds. *Ann Microbiol* 2020;70(1):69. http://doi.org/10.1186/s13213-020-01612-6
140. Database of Physical Properties of Food. http://www.nelfood.com Aaccessed May 10, 2021.
141. 141 European Commission Sustainable Consumption and Production Sustainable Food. https://ec.europa.eu/environment/eussd/food.htm Aaccessed May 12, 2021.
142. Guiné RPF, Bartkiene E, Florença SG, et al. Environmental Issues as Drivers for Food Choice: Study from a Multinational Framework. *Sustainability* 2021;13(5):2869. http://doi.org/10.3390/su13052869
143. Sachs J, Schmidt-Traub G, Kroll C, Lafortune G, Fuller G, Woelm F. *The Sustainable Development Goals and COVID-19*. Sustainable Development Report. Cambridge: Cambridge University Press, 2020.
144. U.S. Department of Agriculture, USDA Agricultural Research Service. FoodData Central. https://fdc.nal.usda.gov Accessed March 10, 2021.
145. Food and Agriculture Organization of the United Nations, FAO, (2021). Food Loss and Waste Database. http://www.fao.org/platform-food-loss-waste/flw-data/en/ Aaccessed May 12, 2021.
146. Poore J, Nemecek T. Reducing food's environmental impacts through producers and consumers. *Science* 2018;360(6392):987–992. http://doi.org/10.1126/science.aaq0216
147. Pakpahan OP, Moreira L, Camelo A, et al. Evaluation of comparative scenarios from different sites of chestnut production using life cycle assessment (LCA): Case study in the Beira Interior region of Portugal. *Heliyon* 2023;9(1):e12847. http://doi.org/10.1016/j.heliyon.2023.e12847
148. van der Werf HMG, Knudsen MT, Cederberg C. Towards better representation of organic agriculture in life cycle assessment. *Nat Sustain* 2020;3(6):419–425. http://doi.org/10.1038/s41893-020-0489-6
149. Durazzo A, Nazhand A, Lucarini M, et al. Occurrence of Tocols in Foods: An Updated Shot of Current Databases. *J Food Qual* 2021;2021:1–7. http://doi.org/10.1155/2021/8857571
150. Guiné RPF, Bartkiene E, Florença SG, et al. Environmental Issues as Drivers for Food Choice: Study from a Multinational Framework. *Sustainability* 2021;13(5):2869. http://doi.org/10.3390/su13052869
151. Tulloch AIT, Oh RRY, Gallegos D. Environmental and public health co-benefits of consumer switches to immunity-supporting food. *Ambio* 2022;51(7):1658-1672. http://doi.org/10.1007/s13280-021-01693-w
152. Ghammachi N, Dharmayani PNA, Mihrshahi S, Ronto R. Investigating Web-Based Nutrition Education Interventions for Promoting Sustainable and Healthy Diets in Young Adults: A Systematic Literature Review. *Int J Environ Res Public Health* 2022;19(3):1691. http://doi.org/10.3390/ijerph19031691
153. Thaiss CA, Zeevi D, Levy M, et al. Transkingdom Control of Microbiota Diurnal Oscillations Promotes Metabolic Homeostasis. *Cell* 2014;159(3):514–529. http://doi.org/10.1016/j.cell.2014.09.048
154. Hall AB, Tolonen AC, Xavier RJ. Human genetic variation and the gut microbiome in disease. *Nat Rev Genet* 2017;18(11):690–699. http://doi.org/10.1038/nrg.2017.63
155. Mahajan MC, McLellan AS. Whole-Exome Sequencing (WES) for Illumina Short Read Sequencers Using Solution-Based Capture. *Methods Mol Biol* 2019:85–108. http://doi.org/10.1007/978-1-4939-9882-1_5
156. Suwinski P, Ong C, Ling MHT, Poh YM, Khan AM, Ong HS. Advancing Personalized Medicine Through the Application of Whole Exome Sequencing and Big Data Analytics. *Front Genet* 2019;10:49. http://doi.org/10.3389/fgene.2019.00049
157. Scherer A, Christensen GB. Concepts and Relevance of Genome-Wide Association Studies. *Sci Prog* 2016;99(1):59–67. http://doi.org/10.3184/003685016X14558068452913

Food and Health 2023;5(1):5. https://doi.org/10.53388/FH2023005

158. Iyer R, Schwartz I. Microarray-Based Comparative Genomic and Transcriptome Analysis of Borrelia burgdorferi. *Microarrays* 2016;5(2):9. http://doi.org/10.3390/microarrays5020009
159. Tong L, Wu P-Y, Phan JH, et al. Impact of RNA-seq data analysis algorithms on gene expression estimation and downstream prediction. *Sci Rep* 2020;10(1):17952. http://doi.org/10.1038/s41598-020-74567-y
160. Cao C, Wang J, Kwok D, et al. webTWAS: a resource for disease candidate susceptibility genes identified by transcriptome-wide association study. *Nucleic Acids Res* 2021;50(D1):D1123–30. http://doi.org/10.1093/nar/gkab957
161. Keseler IM, Gama-Castro S, Mackie A, et al. The EcoCyc Database in 2021. *Front Microbiol* 2021;12. http://doi.org/10.3389/fmicb.2021.711077
162. Kanehisa M, Goto S, Sato Y, Furumichi M, Tanabe M. KEGG for integration and interpretation of large-scale molecular data sets. *Nucleic Acids Res* 2011;40(D1):D109–14. http://doi.org/10.1093/nar/gkr988
163. Jassal B, Matthews L, Viteri G, et al. The reactome pathway knowledgebase. *Nucleic Acids Res* 2020;48(D1):D498–D503. http://doi.org/10.1093/nar/gkz1031
164. Fuess LE, den Haan S, Ling F, Weber JN, Steinel NC, Bolnick DI. Immune Gene Expression Covaries with Gut Microbiome Composition in Stickleback. *mBio* 2021;12(3):e00145-21. http://doi.org/10.1128/mBio.00145-21
165. Fan P, Bian B, Teng L, et al. Host genetic effects upon the early gut microbiota in a bovine model with graduated spectrum of genetic variation. *ISME J* 2019;14(1):302–317. http://doi.org/10.1038/s41396-019-0529-2
166. Eloe-Fadrosh EA, Ahmed F, Babinski M, et al. The national microbiome data collaborative data portal: an integrated multi-omics microbiome data resource. *Nucleic Acids Research* 2022;50(D1):D828–D836. https://academic.oup.com/nar/article/50/D1/D828/6414581?login=true
167. Bharti R, Grimm DG. Current challenges and best-practice protocols for microbiome analysis. *Brief Bioinform* 2019;22(1):178–193. http://doi.org/10.1093/bib/bbz155
168. Shamsaddini A, Dadkhah K, Gillevet PM. BiomMiner: An advanced exploratory microbiome analysis and visualization pipeline. *PLoS ONE* 2020;15(6):e0234860. http://doi.org/10.1371/journal.pone.0234860
169. Liu Y-X, Qin Y, Chen T, et al. A practical guide to amplicon and metagenomic analysis of microbiome data. *Protein Cell* 2020;12(5):315–330. http://doi.org/10.1007/s13238-020-00724-8
170. Khomich M, Måge I, Rud I, Berget I. Analysing microbiome intervention design studies: Comparison of alternative multivariate statistical methods. *PLoS ONE* 2021;16(11):e0259973. http://doi.org/10.1371/journal.pone.0259973
171. Ramon E, Belanche-Muñoz L, Molist F, Quintanilla R, Perez-Enciso M, Ramayo-Caldas Y. kernInt: A Kernel Framework for Integrating Supervised and Unsupervised Analyses in Spatio-Temporal Metagenomic Datasets. *Front Microbiol* 2021;12:609048. http://doi.org/10.3389/fmicb.2021.609048
172. Yakimovich A. Machine Learning and Artificial Intelligence for the Prediction of Host–Pathogen Interactions: A Viral Case. *Infect Drug Resist* 2021;14:3319–3326. http://doi.org/10.2147/IDR.S292743
173. Fiannaca A, La Paglia L, La Rosa M, et al. Deep learning models for bacteria taxonomic classification of metagenomic data. *BMC Bioinformatics* 2018;19(S7):198. http://doi.org/10.1186/s12859-018-2182-6
174. Bolger AM, Lohse M, Usadel B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 2014;30(15):2114–2120. http://doi.org/10.1093/bioinformatics/btu170
175. Bragg L, Stone G, Imelfort M, Hugenholtz P, Tyson GW. Fast, accurate error-correction of amplicon pyrosequences using Acacia. *Nat Methods* 2012;9(5):425–426. http://doi.org/10.1038/nmeth.1990
176. Edgar RC. UNOISE2: improved error-correction for Illumina 16S and ITS amplicon sequencing. *bioRxiv* 2016. https://doi.org/10.1101/081257
177. Edgar RC, Haas BJ, Clemente JC, Quince C, Knight R. UCHIME improves sensitivity and speed of chimera detection. *Bioinformatics* 2011;27(16):2194–2200. http://doi.org/10.1093/bioinformatics/btr381
178. Amir A, McDonald D, Navas-Molina JA,et al. Deblur Rapidly Resolves Single-Nucleotide Community Sequence Patterns. *mSystems* 2017;2(2):e0191-16. http://doi.org/10.1128/mSystems.00191-16
179. Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJA, Holmes SP. DADA2: High-resolution sample inference from Illumina amplicon data. *Nat Methods* 2016;13(7):581–83. http://doi.org/10.1038/nmeth.3869
180. Troll M, Brandmaier S, Reitmeier S, et al. Investigation of Adiposity Measures and Operational Taxonomic unit (OTU) Data Transformation Procedures in Stool Samples from a German Cohort Study Using Machine Learning Algorithms. *Microorganisms* 2020;10;8(4):547. https://doi.org/10.3390/microorganisms8040547
181. Edgar RC. UPARSE: highly accurate OTU sequences from microbial amplicon reads. *Nat Methods* 2013;10(10):996–998. http://doi.org/10.1038/nmeth.2604
182. Wood DE, Salzberg SL. Kraken: ultrafast metagenomic sequence classification using exact alignments. *Genome Biol* 2014;15(3):R46. http://doi.org/10.1186/gb-2014-15-3-r46
183. Shah N, Molloy EK, Pop M, Warnow T. TIPP2: metagenomic taxonomic profiling using phylogenetic markers. *Bioinformatics* 2021;37(13):1839–1845. http://doi.org/10.1093/bioinformatics/btab023
184. Franz M, Whyte L, Atwood TC, et al. Distinct gut microbiomes in two polar bear subpopulations inhabiting different sea ice ecoregions. *Sci Rep* 2022;12(1):522. http://doi.org/10.1038/s41598-021-04340-2
185. Kultima JR, Coelho LP, Forslund K, et al. MOCAT2: a metagenomic assembly, annotation and profiling framework. *Bioinformatics* 2016;32(16):2520–2523. http://doi.org/10.1093/bioinformatics/btw183
186. Zhu J, Li H, Jing ZZ, et al. Robust host source tracking building on the divergent and non-stochastic assembly of gut microbiomes in wild and farmed large yellow croaker. *Microbiome* 2022;10(1):18. http://doi.org/10.1186/s40168-021-01214-7
187. Djemiel C, Maron P-A, Terrat S, Dequiedt S, Cottin A, Ranjard L. Inferring microbiota functions from taxonomic genes: a review. *GigaScience* 2022;11:giab090. http://doi.org/10.1093/gigascience/giab090
188. Casimiro-Soriguer CS, Loucera C, Peña-Chilet M, Dopazo J. Towards a metagenomics machine learning interpretable model for understanding the transition from adenoma to colorectal cancer. *Sci Rep* 2022;12(1):450. http://doi.org/10.1038/s41598-021-04182-y
189. Moreno-Indias I, Lahti L, Nedyalkova M, et al. Statistical and Machine Learning Techniques in Human Microbiome Studies: Contemporary Challenges and Solutions. *Front Microbiol* 2021;12:635781. https://doi.org/10.3389/fmicb.2021.635781
190. Chen Y, Wu T, Lu W, et al. Predicting the Role of the Human Gut Microbiome in Constipation Using Machine-Learning Methods: A Meta-Analysis. *Microorganisms* 2021;9(10):2149. http://doi.org/10.3390/microorganisms9102149

191. Wirbel J, Zych K, Essex M, et al. Microbiome meta-analysis and cross-disease comparison enabled by the SIAMCAT machine learning toolbox. *Genome Biol* 2021;22(1):93. http://doi.org/10.1186/s13059-021-02306-1
192. Karlicki M, Antonowicz S, Karnkowska A. Tiara: deep learning-based classification system for eukaryotic sequences. *Bioinformatics* 2021;38 (2):344–350. http://doi.org/10.1093/bioinformatics/btab672
193. Gordon-Rodriguez E, Quinn TP, Cunningham JP. Learning sparse log-ratios for high-throughput sequencing data. Luigi *Bioinformatics* 2021;38(1):157–163. http://doi.org/10.1093/bioinformatics/btab645
194. Cameron ES, Schmidt PJ, Tremblay BJ-M, Emelko MB, Müller KM. Enhancing diversity analysis by repeatedly rarefying next generation sequencing data describing microbial communities. *Sci Rep* 2021;11(1):22302. http://doi.org/10.1038/s41598-021-01636-1
195. Hird SM. Microbiomes, Community Ecology, and the Comparative Method. *mSystems* 2019;4(3):e00112-19. http://doi.org/10.1128/mSystems.00112-19
196. Alberdi A, Martin Bideguren G, Aizpurua O. Diversity and compositional changes in the gut microbiota of wild and captive vertebrates: a meta-analysis. *Sci Rep* 2021;11(1):22660. http://doi.org/10.1038/s41598-021-02015-6
197. Ortiz-Estrada ÁM, Gollas-Galván T, Martínez-Córdova LR, Martínez-Porchas M. Predictive functional profiles using metagenomic 16S rRNA data: a novel approach to understanding the microbial ecology of aquaculture systems. *Rev Aquacult* 2018;11(1):234–245. http://doi.org/10.1111/raq.12237
198. Kontonatsios G, Brockmeier AJ, Przybyła P, et al. A semi-supervised approach using label propagation to support citation screening. *J Biomed Inform* 2017;72:67–76. http://doi.org/10.1016/j.jbi.2017.06.018
199. Laparra E, Mascio A, Velupillai S, Miller T. A Review of Recent Work in Transfer Learning and Domain Adaptation for Natural Language Processing of Electronic Health Records. *Yearb Med Inform* 2021;30(01):239–244. http://doi.org/10.1055/s-0041-1726522
200. Xiao G, Liu S, Xiao Y, et al. Seasonal Changes in Gut Microbiota Diversity and Composition in the Greater Horseshoe Bat. *Front Microbiol* 2019;10:2247. http://doi.org/10.3389/fmicb.2019.02247
201. Nikolova VL, Smith MRB, Hall LJ, Cleare AJ, Stone JM, Young AH. Perturbations in Gut Microbiota Composition in Psychiatric Disorders. *JAMA Psychiatry* 2021;78(12):1343. http://doi.org/10.1001/jamapsychiatry.2021.2573
202. Korthauer K, Kimes PK, Duvallet C, et al. A practical guide to methods controlling false discoveries in computational biology. *Genome Biol* 2019;20(1):118. http://doi.org/10.1186/s13059-019-1716-1
203. Giacalone M, Agata Z, Cozzucoli PC, Alibrandi A. Bonferroni-Holm and permutation tests to compare health data: methodological and applicative issues. *BMC Med Res Methodol* 2018;18(1):81. http://doi.org/10.1186/s12874-018-0540-8
204. Allen M. *The SAGE encyclopedia of communication research methods (Vols. 1-4)*. Thousand Oaks, CA: SAGE Publications, Inc, 2017.
205. Dobler D, Friedrich S, Pauly M. Nonparametric MANOVA in meaningful effects. *Ann Inst Stat Math* 2019;72(4):997–1022. http://doi.org/10.1007/s10463-019-00717-3
206. Paliy O, Shankar V. Application of multivariate statistical techniques in microbial ecology. *Mol Ecol* 2016;25(5):1032–1057. http://doi.org/10.1111/mec.13536
207. Prochazkova M, Budinska E, Kuzma M, et al. Vegan Diet Is Associated With Favorable Effects on the Metabolic Performance of Intestinal Microbiota: A Cross-Sectional Multi-Omics Study. *Front Nutr*. 2022;8:783302. https://doi.org/10.3389/fnut.2021.783302
208. Tokuhara D. Role of the Gut Microbiota in Regulating Non-alcoholic Fatty Liver Disease in Children and Adolescents. *Front Nutr* 2021;8:700058. http://doi.org/10.3389/fnut.2021.700058
209. Metwaly A, Dunkel A, Waldschmitt N, et al. Integrated microbiota and metabolite profiles link Crohn's disease to sulfur metabolism. *Nat Commun* 2020;11(1):4322. http://doi.org/10.1038/s41467-020-17956-1
210. William Sherwin. Is Bray-Curtis differentiation meaningful in Molecular Ecology? *Authorea* 2021. https://doi.org/10.22541/au.161839260.09775220/v1
211. Suzuki K, Abe MS, Kumakura D, et al. Chemical-Mediated Microbial Interactions Can Reduce the Effectiveness of Time-Series-Based Inference of Ecological Interaction Networks. *Int J Environ Res Public Health* 2022;19(3):1228. http://doi.org/10.3390/ijerph19031228
212. Gat D, Reicher N, Schechter S, et al. Size-Resolved Community Structure of Bacteria and Fungi Transported by Dust in the Middle East. *Front Microbiol* 2021;12:744117. http://doi.org/10.3389/fmicb.2021.744117
213. Hirano H, Takemoto K. Difficulty in inferring microbial community structure based on co-occurrence network approaches. *BMC Bioinformatics* 2019;20(1):329. http://doi.org/10.1186/s12859-019-2915-1
214. Alshawaqfeh M, Serpedin E, Younes AB. Inferring microbial interaction networks from metagenomic data using SgLV-EKF algorithm. *BMC Genomics* 2017;18(S3):228. http://doi.org/10.1186/s12864-017-3605-x
215. Verdi S, Abbasian G, Bowyer RCE, et al. TwinsUK: The UK Adult Twin Registry Update. *Twin Res Hum Genet* 2019;22(6):523–529. http://doi.org/10.1017/thg.2019.65
216. Devlin SM, Martin A, Ostrovnaya I. Identifying prognostic pairwise relationships among bacterial species in microbiome studies. *PLoS Comput Biol* 2021;17(11):e1009501. http://doi.org/10.1371/journal.pcbi.1009501
217. George S, Aguilera X, Gallardo P, et al. Bacterial Gut Microbiota and Infections During Early Childhood. *Front Microbiol* 2022;12:793050. http://doi.org/10.3389/fmicb.2021.793050
218. Lima-Filho LMA, Pereira TL, Souza TC, Bayer FM. Process monitoring using inflated beta regression control chart. *PLoS ONE* 2020;15(7):e0236756. http://doi.org/10.1371/journal.pone.0236756
219. Mallick H, Rahnavard A, McIver LJ, et al. Multivariable association discovery in population-scale meta-omics studies. *PLoS Comput Biol* 2021;17(11):e1009442. http://doi.org/10.1371/journal.pcbi.1009442
220. Ling W, Zhao N, Plantinga AM, et al. Powerful and robust non-parametric association testing for microbiome data via a zero-inflated quantile approach (ZINQ). *Microbiome* 2021;9(1):181. http://doi.org/10.1186/s40168-021-01129-3
221. Bien J, Yan X, Simpson L, Müller CL. Tree-aggregated predictive modeling of microbiome data. *Sci Rep* 2021;11(1):14505. http://doi.org/10.1038/s41598-021-93645-3
222. Ince RAA, Giordano BL, Kayser C, Rousselet GA, Gross J, Schyns PG. A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula. *Hum Brain Mapp* 2016;38(3):1541–1573. http://doi.org/10.1002/hbm.23471
223. Wang W, Kong W, Wang S, Wei K. Detecting Biomarkers of Alzheimer's Disease Based on Multi-constrained Uncertainty-Aware Adaptive Sparse Multi-view Canonical Correlation Analysis. *J Mol Neurosci* 2022;72(4):841–865. http://doi.org/10.1007/s12031-021-01963-y
224. Li S, Kararigas G. Role of Biological Sex in the

Cardiovascular-Gut Microbiome Axis. *Front Cardiovasc Med* 2022;8:759735.
http://doi.org/10.3389/fcvm.2021.759735
225. Lu S, Shi X, Li M, Jiao J, Feng L, Wang G. Semi-supervised random forest regression model based on co-training and grouping with information entropy for evaluation of depression symptoms severity. *Math Biosci Eng* 2021;18(4):4586–4602.
http://doi.org/10.3934/mbe.2021233
226. Jiang T, Gradus JL, Rosellini AJ. Supervised Machine Learning: A Brief Primer. *Behav Ther* 2020;51(5):675–687.
http://doi.org/10.1016/j.beth.2020.05.002
227. Tkach S, Dorofeyev A, Kuzenko I, et al. Current Status and Future Therapeutic Options for Fecal Microbiota Transplantation. *Medicina (Mex)* 2022;58(1):84.
http://doi.org/10.3390/medicina58010084
228. Huang S, Cai N, Pacheco PP, Narrandes S, Wang Y, Xu W. Applications of Support Vector Machine (SVM) Learning in Cancer Genomics. *Cancer Genomics Proteomics* 2018;15(1):41–51.
http://doi.org/10.21873/cgp.20063
229. Gu B, Xiong Z, Li X, Zhai Z, Zheng G. Kernel Path for ν-Support Vector Classification. *IEEE Trans Neural Netw Learn Syst* 2021;34(1):PP.
https://doi.org/10.1109/tnnls.2021.3097248
230. Guo Y, Feng Y, Qu F, Zhang L, Yan B, Lv J. Prediction of hepatitis E using machine learning models. *PLoS ONE* 2020;15(9):e0237750.
http://doi.org/10.1371/journal.pone.0237750
231. Sahu A, Blätke M-A, Szymański JJ, Töpfer N. Advances in flux balance analysis by integrating machine learning and mechanism-based models. *Comput Struct Biotechnol J* 2021;19:4626–4640.
http://doi.org/10.1016/j.csbj.2021.08.004
232. Arslan E, Schulz J, Rai K. Machine Learning in Epigenomics: Insights into Cancer Biology and Medicine. *Biochim Biophys Acta Rev Cancer* 2021;1876(2):188588.
http://doi.org/10.1016/j.bbcan.2021.188588
233. Lin CH, Lichtarge O. Using interpretable deep learning to model cancer dependencies. *Bioinformatics* 2021;37(17):2675-2681.
http://doi.org/10.1093/bioinformatics/btab137
234. Jiang Y, Yang M, Wang S, Li X, Sun Y. Emerging role of deep learning-based artificial intelligence in tumor pathology. *Cancer Commun* 2020;40(4):154–166.
http://doi.org/10.1002/cac2.12012
235. Chen Z, Lin L, Wu C, Li C, Xu R, Sun Y. Artificial intelligence for assisting cancer diagnosis and treatment in the era of precision medicine. *Cancer Commun* 2021;41(11):1100–1115.
http://doi.org/10.1002/cac2.12215
236. Sarker IH. Deep Learning: A Comprehensive Overview on Techniques, Taxonomy, Applications and Research Directions. *SN COMPUT SCI* 2021;2(6):420.
http://doi.org/10.1007/s42979-021-00815-1
237. Vigna L, Silvia Tirelli A, Grossi E, et al. Directional Relationship Between Vitamin D Status and Prediabetes: A New Approach from Artificial Neural Network in a Cohort of Workers with Overweight-Obesity. *J Am Coll Nutr* 2019;38(8):681–692.
http://doi.org/10.1080/07315724.2019.1590249
238. Salvatore S, Bramness JG, Røislien J. Erratum to: Exploring functional data analysis and wavelet principal component analysis on ecstasy (MDMA) wastewater data. *BMC Med Res Methodol* 2017;17(1):34.
http://doi.org/10.1186/s12874-017-0311-y
239. Sisk-Hackworth L, Kelley ST. An application of compositional data analysis to multiomic time-series data. *NAR Genomics and Bioinformatics* 2020;2(4):lqaa079.
http://doi.org/10.1093/nargab/lqaa079
240. Leong C, Haszard JJ, Heath A-LM, et al. Using compositional principal component analysis to describe children's gut microbiota in relation to diet and body composition. *Am J Clin Nutr* 2020;111(1):70–78.
http://doi.org/10.1093/ajcn/nqz270
241. Noh H, Jang H-H, Kim G, et al. Taxonomic Composition and Diversity of the Gut Microbiota in Relation to Habitual Dietary Intake in Korean Adults. *Nutrients* 2021;13(2):366.
http://doi.org/10.3390/nu13020366
242. Wang X, Xiao Y, Xu X, et al. Characteristics of Fecal Microbiota and Machine Learning Strategy for Fecal Invasive Biomarkers in Pediatric Inflammatory Bowel Disease. *Front Cell Infect Microbiol* 2021;11:711884.
http://doi.org/10.3389/fcimb.2021.711884
243. Rahman SF, Olm MR, Morowitz MJ, Banfield JF. Machine Learning Leveraging Genomes from Metagenomes Identifies Influential Antibiotic Resistance Genes in the Infant Gut Microbiome. *mSystems* 2018;3(1):e00123-17.
http://doi.org/10.1128/mSystems.00123-17
244. Melo MCR, Maasch JRMA, de la Fuente-Nunez C. Accelerating antibiotic discovery through artificial intelligence. *Commun Biol* 2021;4(1):1050.
http://doi.org/10.1038/s42003-021-02586-0
245. Yang X, Wang Y, Byrne R, Schneider G, Yang S. Concepts of Artificial Intelligence for Computer-Assisted Drug Discovery. *Chem Rev* 2019;119(18):10520–10594.
http://doi.org/10.1021/acs.chemrev.8b00728
246. Hulot A, Laloë D, Jaffrézic F. A unified framework for the integration of multiple hierarchical clusterings or networks from multi-source data. *BMC Bioinformatics* 2021;22(1):392.
http://doi.org/10.1186/s12859-021-04303-4
247. Robertson RC, Kaliannan K, Strain CR, Ross RP, Stanton C, Kang JX. Maternal omega-3 fatty acids regulate offspring obesity through persistent modulation of gut microbiota. *Microbiome* 2018;6(1):95.
http://doi.org/10.1186/s40168-018-0476-6
248. Alabduljabbar S, Zaidan SA, Lakshmanan AP, Terranegra A. Personalized Nutrition Approach in Pregnancy and Early Life to Tackle Childhood and Adult Non-Communicable Diseases. *Life* 2021;11(6):467.
http://doi.org/10.3390/life11060467
249. Fleming JS. Explorer: A program for common factor analysis and related models for data analysis [Computer software], 2017.
250. Yong AG, Pearce S. A Beginner's Guide to Factor Analysis: Focusing on Exploratory Factor Analysis. *Tutor Quant. Methods Psychol* 2013;9(2):79–94.
http://doi.org/10.20982/tqmp.09.2.p079
251. Buffet-Bataillon S, Bellanger A, Boudry G, et al. New Insights Into Microbiota Modulation-Based Nutritional Interventions for Neurodevelopmental Outcomes in Preterm Infants. *Front Microbiol* 2021;12:676622.
http://doi.org/10.3389/fmicb.2021.676622
252. Chen Z, Shi K, Liu X, et al. Gut Microbial Profile Is Associated With the Severity of Social Impairment and IQ Performance in Children With Autism Spectrum Disorder. *Front Psychiatry* 2021;12:789864.
http://doi.org/10.3389/fpsyt.2021.789864
253. Gomez R, Stavropoulos V. Confirmatory Factor Analysis and Exploratory Structural Equation Modeling of the Structure of Attention-Deficit/Hyperactivity Disorder Symptoms in Adults. *Assessment* 2020;28(6):1570–1582.
http://doi.org/10.1177/1073191120905892
254. Wang S, Wei Y, Liu L, Li Z. Association Between Breastmilk Microbiota and Food Allergy in Infants. *Front Cell Infect Microbiol* 2022;11:770913.
http://doi.org/10.3389/fcimb.2021.770913
255. Moossavi S, Sepehri S, Robertson B, et al. Composition and Variation of the Human Milk Microbiota Are Influenced by Maternal and Early-Life Factors. *Cell Host Microbe* 2019;25(2):324-335.e4.
http://doi.org/10.1016/j.chom.2019.01.011

256. Ray-Offor E, Amadi S. Hemorrhoidal disease: Predilection sites, pattern of presentation, and treatment. *Ann Afr Med* 2019;18(1):12. http://doi.org/10.4103/aam.aam_4_18
257. Hallquist MN, Wiley JF. MplusAutomation: An R Package for Facilitating Large-Scale Latent Variable Analyses in Mplus. *Struct Equ Modeling* 2018;25(4):621–638. http://doi.org/10.1080/10705511.2017.1402334
258. Mead A. Review of the Development of Multidimensional Scaling Methods. *The Statistician* 1992;41(1):27–39. http://doi.org/10.2307/2348634
259. Lin A, Bik EM, Costello EK, et al. Distinct Distal Gut Microbiome Diversity and Composition in Healthy Children from Bangladesh and the United States. *PLoS ONE* 2013;8(1):e53838. http://doi.org/10.1371/journal.pone.0053838
260. Vitali F, Tortora K, Di Paola M, et al. Intestinal microbiota profiles in a genetic model of colon tumorigenesis correlates with colon cancer biomarkers. *Sci Rep* 2022;12(1):1432. http://doi.org/10.1038/s41598-022-05249-0
261. Laios A, De Oliveira Silva RV, Dantas De Freitas DL, et al. Machine Learning-Based Risk Prediction of Critical Care Unit Admission for Advanced Stage High Grade Serous Ovarian Cancer Patients Undergoing Cytoreductive Surgery: The Leeds-Natal Score. *J Clin Med* 2021;11(1):87. http://doi.org/10.3390/jcm11010087
262. Way GP, Zietz M, Rubinetti V, Himmelstein DS, Greene CS. Compressing gene expression data using multiple latent space dimensionalities learns complementary biological representations. *Genome Biol* 2020;21(1):109. http://doi.org/10.1186/s13059-020-02021-3
263. Malmuthuge N, Liang G, Griebel PJ, Guan LL. Taxonomic and Functional Compositions of the Small Intestinal Microbiome in Neonatal Calves Provide a Framework for Understanding Early Life Gut Health. *Appl Environ Microbiol* 2019;85(6):e02534-18. http://doi.org/10.1128/AEM.02534-18
264. Martinez-Medina JN, Flores-Lopez R, López-Contreras BE, et al. Effect of Gut Microbial Enterotypes on the Association between Habitual Dietary Fiber Intake and Insulin Resistance Markers in Mexican Children and Adults. *Nutrients* 2021;13(11):3892. http://doi.org/10.3390/nu13113892
265. Johnson MG, Pokorny L, Dodsworth S, et al. A Universal Probe Set for Targeted Sequencing of 353 Nuclear Genes from Any Flowering Plant Designed Using k-Medoids Clustering. Renner S, editor. *Syst Biol* 2018;68(4):594–606. http://doi.org/10.1093/sysbio/syy086
266. Karim MR, Beyan O, Zappa A, et al. Deep learning-based clustering approaches for bioinformatics. *Brief Bioinform* 2020;22(1):393–415. http://doi.org/10.1093/bib/bbz170
267. Handl J, Knowles J, Kell DB. Computational cluster validation in post-genomic data analysis. *Bioinformatics* 2005;21(15):3201–3212. http://doi.org/10.1093/bioinformatics/bti517
268. McElrath TF, Cantonwine DE, Gray KJ, et al. Late first trimester circulating microparticle proteins predict the risk of preeclampsia < 35 weeks and suggest phenotypic differences among affected cases. *Sci Rep* 2020;10(1):17353. http://doi.org/10.1038/s41598-020-74078-w
269. Zhang J, Wang P, Dingkao R, et al. Fecal Microbiota Dynamics Reveal the Feasibility of Early Weaning of Yak Calves under Conventional Grazing System. *Biology* 2021;11(1):31. http://doi.org/10.3390/biology11010031
270. Piening BD, Zhou W, Contrepois K, et al. Integrative personal omics profiles during periods of weight gain and loss. *Cell Syst* 2018;6(2):157–170. http://doi.org/10.1016/j.cels.2017.12.013.
271. Nicora G, Vitali F, Dagliati A, Geifman N, Bellazzi R. Integrated multi-omics analyses in oncology: a review of machine learning methods and tools. *Front Oncol* 2020;10. http://doi.org/10.3389/fonc.2020.01030
272. Prochazkova M, Budinska E, Kuzma M, et al. Vegan diet is associated with favorable effects on the metabolic performance of intestinal microbiota: a cross-sectional multi-omics study. *Front Nutr* 2022;8:783302. http://doi.org/10.3389/fnut.2021.783302
273. Nolte Fong JV, Miketinas D, Moore LW, et al. Precision nutrition model predicts glucose control of overweight females following the consumption of potatoes high in resistant starch. *Nutrients* 2022;14(2):268. http://doi.org/10.3390/nu14020268
274. Sorbara MT, Pamer EG. Microbiome-based therapeutics. *Nat Rev Microbiol* 2022;20(6):365–380. http://doi.org/10.1038/s41579-021-00667-9
275. Dimet-Wiley A, Wu Q, Wiley JT, et al. Reduced calorie diet combined with NNMT inhibition establishes a distinct microbiome in DIO mice. *Sci Rep* 2022;12(1):484. http://doi.org/10.1038/s41598-021-03670-5
276. Verburgt CM, Ghiboub M, Benninga MA, de Jonge WJ, Van Limbergen JE. Nutritional therapy strategies in pediatric crohn's disease. *Nutrients* 2021;13(1):212. http://doi.org/10.3390/nu13010212
277. Saa P, Urrutia A, Silva-Andrade C, Martín AJ, Garrido D. Modeling approaches for probing cross-feeding interactions in the human gut microbiome. *Comput Struct Biotechnol J* 2022;20:79–89. http://doi.org/10.1016/j.csbj.2021.12.006
278. Altamirano Á, Saa PA, Garrido D. Inferring composition and function of the human gut microbiome in time and space: A review of genome-scale metabolic modelling tools. *Comput Struct Biotechnol J* 2020;18:3897–3904. http://doi.org/10.1016/j.csbj.2020.11.035
279. Jean-Pierre F, Henson MA, O'Toole GA. Metabolic modeling to interrogate microbial disease: a tale for experimentalists. *Front Mol Biosci* 2021;8. http://doi.org/10.3389/fmolb.2021.634479
280. Xue Y, Thalmayer AS, Zeising S, Fischer G, Lübke M. Commercial and scientific solutions for blood glucose monitoring—a review. *Sensors* 2022;22(2):425. http://doi.org/10.3390/s22020425
281. De Filippis F, Esposito A, Ercolini D. Outlook on next-generation probiotics from the human gut. *Cell Mol Life Sci* 2022;79(2):76. http://doi.org/10.1007/s00018-021-04080-6
282. Rizzello CG, Portincasa P, Montemurro M, et al. Sourdough fermented breads are more digestible than those started with baker's yeast alone: an in vivo challenge dissecting distinct gastrointestinal responses. *Nutrients* 2019;11(12):2954. http://doi.org/10.3390/nu11122954
283. Da Ros A, Polo A, Rizzello CG, et al. Feeding with sustainably sourdough bread has the potential to promote the healthy microbiota metabolism at the colon level. *Microbiol Spectr* 2021;9(3):e0049421. http://doi.org/10.1128/Spectrum.00494-21
284. Jamilu Ibrahim A, Zira P, Abdulganiyyi N. Hybrid recommender for research papers and articles. *IJIIS* 2021;10(2):9. http://doi.org/10.11648/j.ijiis.20211002.11
285. Wang Y, Gao X, Ghozlane A, et al. Characteristics of faecal microbiota in paediatric crohn's disease and their dynamic changes during infliximab therapy. *J Crohns Colitis* 2017;12(3):337–346. http://doi.org/10.1093/ecco-jcc/jjx153
286. Schmidt CM, Ghadermazi P, Chan SHJ. Predicting microbiome metabolism and interactions through integrating multidisciplinary principles. *mSystems* 2021;6(5):e0076821. http://doi.org/10.1128/mSystems.00768-21
287. Sauter T, Bintener T, Kishk A, et al. Project-based learning course on metabolic network modelling in computational systems biology. *PLoS Comput Biol* 2022;18(1):e1009711. http://doi.org/10.1371/journal.pcbi.1009711

288. Noronha A, Modamio J, Jarosz Y, et al. The Virtual Metabolic Human database: integrating human and gut microbiome metabolism with nutrition and disease. *Nucleic Acids Res* 2018;47(D1):D614–D624. http://doi.org/10.1093/nar/gky992
289. Lin D, Zheng X, Sanogo B, Ding T, Sun X, Wu Z. Bacterial composition of midgut and entire body of laboratory colonies of Aedes aegypti and Aedes albopictus from Southern China. *Parasit Vectors* 2021;14(1):586. http://doi.org/10.1186/s13071-021-05050-4
290. Blasco T, Pérez-Burillo S, Balzerani F, et al. An extended reconstruction of human gut microbiota metabolism of dietary compounds. *Nat Commun* 2021;12(1):4728. http://doi.org/10.1038/s41467-021-25056-x
291. Hinojosa-Nogueira D, Pérez-Burillo S, Navajas-Porras B, et al. Development of an unified food composition database for the European project "Stance4Health." *Nutrients* 2021;13(12):4206. http://doi.org/10.3390/nu13124206
292. Zou Y, Xue W, Luo G, et al. 1,520 reference genomes from cultivated human gut bacteria enable functional microbiome analyses. *Nat Biotechnol* 2019;37(2):179–185. http://doi.org/10.1038/s41587-018-0008-8
293. Bell E, Binkowski S, Sanderson E, et al. Substantial intra-individual variability in post-prandial time to peak in controlled and free-living conditions in children with type 1 diabetes. *Nutrients* 2021;13(11):4154. http://doi.org/10.3390/nu13114154
294. Mills, Lane, Smith, Grimaldi, Ross, Stanton. Precision nutrition and the microbiome part II: potential opportunities and pathways to commercialisation. *Nutrients* 2019;11(7):1468. http://doi.org/10.3390/nu11071468
295. Barone F, Laghi L, Gianotti A, et al. In vivo effects of einkorn wheat (triticum monococcum) bread on the intestinal microbiota, metabolome, and on the glycemic and insulinemic response in the pig model. *Nutrients* 2018;11(1):16. http://doi.org/10.3390/nu11010016
296. Gupta G, Katarya R. Research on understanding the effect of deep learning on user preferences. *Arab J Sci Eng* 2020;46(4):3247–3286. http://doi.org/10.1007/s13369-020-05112-2
297. Buhmann MD, Melville P, Sindhwani V, et al. Recommender systems. *Encycl Machine Learning* 2011:829–838. http://doi.org/10.1007/978-0-387-30164-8_705
298. Ko H, Lee S, Park Y, Choi A. A survey of recommendation systems: recommendation models, techniques, and application fields. *Electronics* 2022;11(1):141. http://doi.org/10.3390/electronics11010141
299. Alili R, Belda E, Fabre O, et al. Characterization of the gut microbiota in individuals with overweight or obesity during a real-world weight loss dietary program: a focus on the bacteroides 2 enterotype. *Biomedicines* 2021;10(1):16. http://doi.org/10.3390/biomedicines10010016
300. Bhalla P, Rengaswamy R, Karunagaran D, Suraishkumar GK, Sahoo S. Metabolic modeling of host–microbe interactions for therapeutics in colorectal cancer. *npj Syst Biol Appl* 2022;8(1):1. http://doi.org/10.1038/s41540-021-00210-9
301. Heirendt L, Arreckx S, Pfau T, et al. Creation and analysis of biochemical constraint-based models using the COBRA Toolbox v.3.0. *Nat Proto*c 2019;14(3):639–702.
302. Prados-Bo A, Casino G. Microbiome research in general and business newspapers: how many microbiome articles are published and which study designs make the news the most? *PLoS One* 2021;16(4):e0249835. http://doi.org/10.1371/journal.pone.0249835
303. Taroncher-Oldenburg G, Jones S, Blaser M, et al. Translating microbiome futures. *Nat Biotechnol* 2018;36(11):1037–1042. http://doi.org/10.1038/nbt.4287
304. Çavdar G, Papich T, Ryan EP. Microbiome, breastfeeding and public health policy in the united states: the case for dietary fiber. *Nutr Metab Insights* 2019;12:117863881986959. http://doi.org/10.1177/1178638819869597